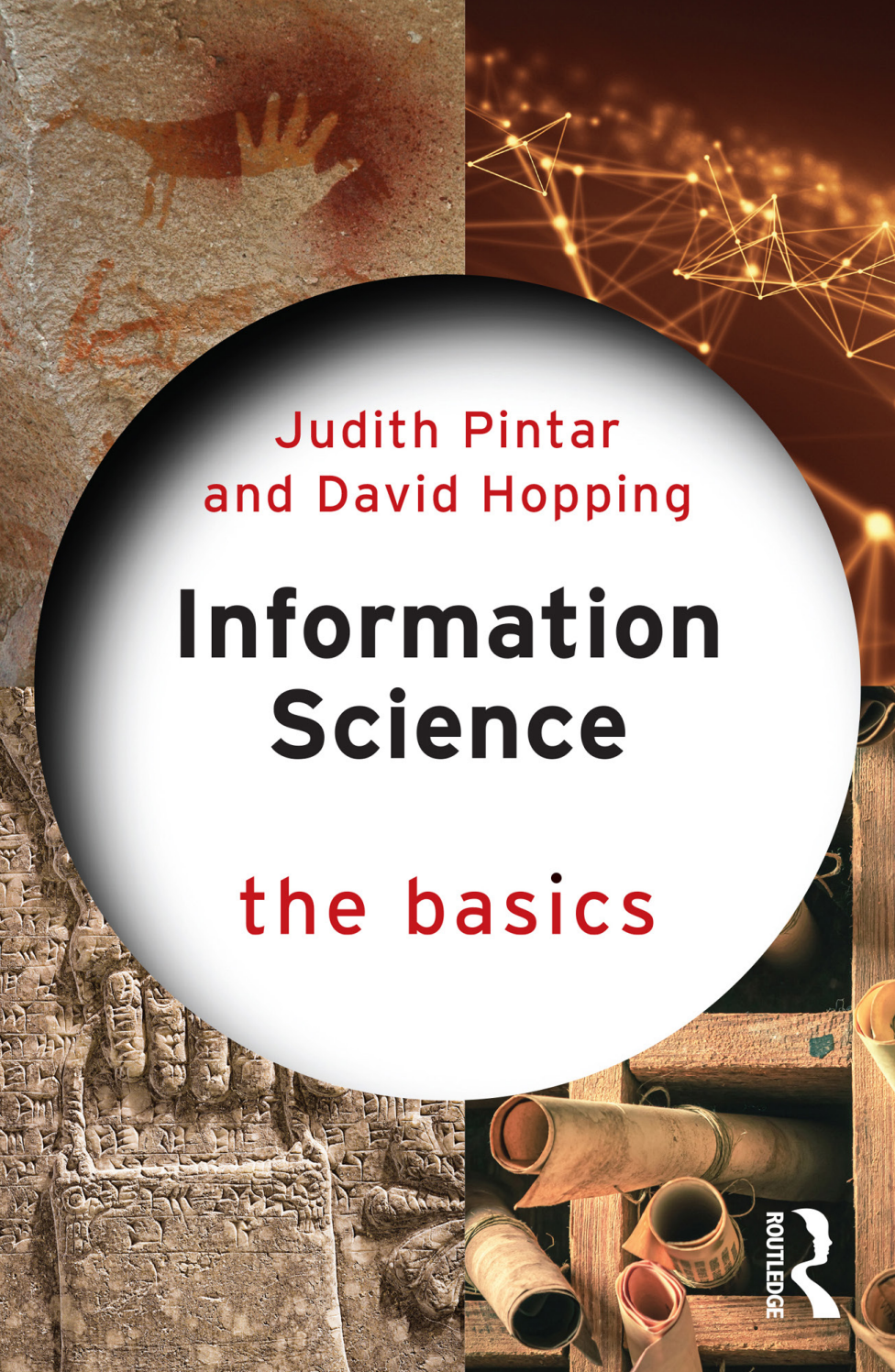


The background of the book cover is a collage of four distinct images. The top-left corner shows a close-up of ancient rock art, featuring a reddish-brown handprint and a stylized animal figure. The top-right corner displays a glowing orange network diagram with interconnected nodes and lines, suggesting a digital or scientific theme. The bottom-left corner depicts a detailed stone relief carving with intricate patterns and figures. The bottom-right corner shows several rolled-up ancient scrolls resting on a wooden surface.

Judith Pinter
and David Hopping

Information Science

the basics

ROUTLEDGE



INFORMATION SCIENCE

Information Science: The Basics provides an accessible introduction to the multifaceted field of Information Science (IS).

Inviting readers to explore a modern field of study with deep historical foundations, the book begins by considering the complexities of the term “information” and the information life cycle from classification to preservation. Each chapter examines a different area within IS, surveying its history, technologies, and practices with a critical eye. This interdisciplinary field incorporates a wide range of approaches which it shares with humanities, social science, and technology fields. What makes IS unique is its emphasis on the connections between information, technology, and society. The need to share information more effectively in response to social, environmental, and biomedical challenges has never been so urgent; the volume discusses the risks as well as benefits that come with the emerging technologies that make it possible. The book also explores how IS, with its long-standing commitment to intellectual freedom and digital inclusion, and its keen attention to the protection of privacy, data ethics, and algorithmic transparency, can contribute to the creation of a more open and equitable society.

Information Science: The Basics is essential reading for anyone who wishes to know more about information and the impact it has on our world. It will be particularly useful for anyone intending to study IS at the undergraduate level or considering a shift to a career in the information professions.

Judith Pintar is a faculty member and directs the Game Studies & Design program in the School of Information Sciences at the University of Illinois Urbana-Champaign. In 2020, she was named a University of Illinois Distinguished Teacher/Scholar. She teaches introduction to information sciences, and interactive narrative design. Her research interests include game studies, gameful pedagogy, digital literacies, immersive storytelling, narrative AI, the digital afterlife, propaganda, persuasion, and disinformation.

David Hopping is a faculty member and Director of Workforce Development/Continuing Education in the School of Information Sciences at the University of Illinois Urbana-Champaign. He teaches web design, information architecture, and web content strategies. His research interests include social and community informatics, sociological theories of human rights, action research and public policy relating to digital inclusion and digital literacy, and biofeedback game design in virtual environments.



Taylor & Francis

Taylor & Francis Group

<http://taylorandfrancis.com>

THE BASICS SERIES

The Basics is a highly successful series of accessible guidebooks which provide an overview of the fundamental principles of a subject area in a jargon-free and undaunting format.

Intended for students approaching a subject for the first time, the books both introduce the essentials of a subject and provide an ideal springboard for further study. With over 50 titles spanning subjects from artificial intelligence (AI) to women's studies, *The Basics* are an ideal starting point for students seeking to understand a subject area.

Each text comes with recommendations for further study and gradually introduces the complexities and nuances within a subject.

WORLD PREHISTORY

BRIAN M. FAGAN AND NADIA DURRANI

FRENCH REVOLUTION

DARIUS VON GÜTTNER

RESEARCH METHODS (THIRD EDITION)

NICHOLAS WALLIMAN

ARCHAEOLOGY (FOURTH EDITION)

BRIAN M. FAGAN AND NADIA DURRANI

REAL ESTATE

JAN WILCOX AND JANE FORSYTH

MANAGEMENT (SECOND EDITION)

MORGEN WITZEL

SEMIOTICS (FOURTH EDITION)

DANIEL CHANDLER

CHOREOGRAPHY

JENNY ROCHE AND STEPHANIE BURRIDGE

LANGUAGE ACQUISITION

PAUL IBBOTSON

AIR POLLUTION AND CLIMATE CHANGE

JOHN PEARSON AND

RICHARD DERWENT

INFANCY

MARC H. BORNSTEIN AND MARTHA E.

ARTERBERRY

EDUCATIONAL NEUROSCIENCE

CATHY ROGERS AND MICHAEL S. C.

THOMAS

CLASSICAL MYTHOLOGY (SECOND EDITION)

RICHARD MARTIN

PLAY DIRECTING: THE BASICS

DAMON KIELY

ARCHAEOLOGICAL THEORY

ROBERT CHAPMAN

INFORMATION SCIENCE

JUDITH PINTAR AND DAVID HOPPING

DEATH AND RELIGION

CANDI K. CANN

DIGITAL RELIGION

HEIDI A. CAMPBELL AND WENDI BELLAR

DRAMATURGY

ANNE M. HAMILTON AND WALTER

BYONGSOK CHON

For a full list of titles in this series, please visit www.routledge.com/The-Basics/book-series/B

INFORMATION SCIENCE

THE BASICS

Judith Pinter and David Hopping

 **Routledge**
Taylor & Francis Group
LONDON AND NEW YORK

Cover image: © Getty Images

First published 2023

by Routledge

4 Park Square, Milton Park, Abingdon, Oxon OX14 4RN

and by Routledge

605 Third Avenue, New York, NY 10158

Routledge is an imprint of the Taylor & Francis Group, an informa business

© 2023 Judith Pintar and David Hopping

The right of Judith Pintar and David Hopping to be identified as authors of this work has been asserted in accordance with sections 77 and 78 of the Copyright, Designs and Patents Act 1988.

All rights reserved. No part of this book may be reprinted or reproduced or utilised in any form or by any electronic, mechanical, or other means, now known or hereafter invented, including photocopying and recording, or in any information storage or retrieval system, without permission in writing from the publishers.

Trademark notice: Product or corporate names may be trademarks or registered trademarks, and are used only for identification and explanation without intent to infringe.

British Library Cataloguing-in-Publication Data

A catalogue record for this book is available from the British Library

Library of Congress Cataloging-in-Publication Data

Names: Pintar, Judith, author. | Hopping, David, author.

Title: Information science : the basics / Judith Pintar and David Hopping.

Description: Milton Park, Abingdon, Oxon ; New York, NY : Routledge, 2023. |

Series: The basics | Includes bibliographical references and index. |

Identifiers: LCCN 2022042213 (print) | LCCN 2022042214 (ebook) |

ISBN 9780367725204 (hardback) | ISBN 9780367725181 (paperback) |

ISBN 9781003155119 (ebook)

Subjects: LCSH: Information science.

Classification: LCC Z665 P56 2022 (print) | LCC Z665 (ebook) |

DDC 020--dc23/eng/20221020

LC record available at <https://lcn.loc.gov/2022042213>

LC ebook record available at <https://lcn.loc.gov/2022042214>

ISBN: 978-0-367-72520-4 (hbk)

ISBN: 978-0-367-72518-1 (pbk)

ISBN: 978-1-003-15511-9 (ebk)

DOI: 10.4324/9781003155119

Typeset in Bembo

by Deanta Global Publishing Services, Chennai, India

CONTENTS

<i>Preface</i>	<i>viii</i>
<i>Introduction</i>	<i>x</i>
1 Collecting Information	1
2 Classifying Information	22
3 Organizing Information	43
4 Retrieving Information	64
5 Utilizing Information	86
6 Governing Information	108
7 Studying Information	130
8 Designing Information	152
9 Preserving Information	173
10 Imagining Information	194
<i>Bibliography</i>	<i>205</i>
<i>Index</i>	<i>238</i>

PREFACE

Information Science: The Basics provides an accessible introduction to a complex field. Some readers may (justly) feel that important topics were left out, or that others do not belong in the chapters where they were placed. If a history book is a sonnet, a basics book is a haiku! We have chosen to take a narrative approach to the material. We tell stories about historical figures in the text, but credit most contemporary scholars, even luminaries, in the endnotes unless they are the subject of the tale being told. We encourage our readers to follow the trails suggested by the notes to learn more about the people whose ideas spark their curiosity.

We are fortunate that information science is rich with histories, textbooks, and encyclopedias which trace the professional accomplishments of important scholars, map the controversies, and provide technical details of the tools and practices of information science in a more detailed and comprehensive way than we can do here. We direct readers, in particular, to the work of Marcia Bates, Michael Buckland, David Bawden, Lyn Robinson, and William Aspray whose long arc perspectives on the history of information science illuminate the field's complexity and promise.

We are grateful to our colleagues Lisa Bievenue, Maria Bonn, Lisa Hinchliffe, Kathryn LaBarre, Emily Knox, Melissa Ocepek, Madelyn Sanfilippo, Jodi Schneider, Linda Smith, and Mike Twidale for their moral support and editorial suggestions. We also

wish to acknowledge our anonymous reviewers whose productive comments narrowed our narrative at key points, and broadened it at others, with special thanks to the one who will immediately recognize their contribution to the introduction.

The book is dedicated to the memory of Jerome “Jerry” McDonough (1963–2021) who never let us forget that information science is a network of relations. Both consciousness and responsibility are a requirement of that belonging:

The technological web and the social web cannot be regarded as separate. The threads of technology cannot be rewoven without altering the social web, and vice versa. If designers wish to build a better world, they will need to fully comprehend the webs which they weave, and those in which they are entangled.¹

NOTE

- 1 Jerome Patrick McDonough, *Under Construction: The Application of a Feminist Sociology to Information Systems Design* (Berkeley, CA: University of California, 2000): 228–229.

INTRODUCTION

In the first months of the Covid-19 pandemic, schools across the world found themselves suddenly faced with quarantine conditions. Still needing to serve their students, they hastened to put their classrooms online. A photograph captured that moment and its dramatic contradictions: two young girls sit cross-legged on a public sidewalk outside of a Taco Bell restaurant in Salinas California, using its public WiFi connection in order to attend school.¹ An impressive array of technologies had made it possible to transition rapidly from physical classrooms to online learning spaces, but it was also clear that the benefits of those technologies were unequally distributed. The educational experiences of some students, and their ability to access the information they needed during the early months of the pandemic, were dramatically different from the experiences of other young people, located in different geographic and socioeconomic places across town and across the globe.

Two students sit outside a Taco Bell to use Wi-Fi so they can “go to school” online.

This is California, home to Silicon Valley...but where the digital divide is as deep as ever.

Where 40% of all Latinos don't have internet access. This generation deserves better.²

As public sympathy poured out towards the family, commentators noted that the problem was actually widespread. California Senator

Kevin de León tweeted the viral photo, noting that 40% of Latino families in the state lacked access to the Internet.³

The Salinas City Elementary School district subsequently provided their students with 1,500 WiFi hotspots, along with 10,000 Chromebooks. The girls' case was used to spearhead California Senate Bill 156, "Broadband for All," which passed the following year. The legislation allocated six billion dollars for broadband infrastructure in California, noting that two million Californians were without high-speed internet access, including 50% of rural housing units.⁴

Uneven distribution of *information and communication technologies* (ICTs), including access to the Internet through local infrastructure, and to the devices able to make use of high-speed connections, reflects other information stratifications that exist within and between nations. The sociotechnical issues associated with the digital divide are only one part of a more complicated set of global information challenges. Even when people have access to the Internet, not everything that they find through their computers, phones, and other devices is accurate, nor is it necessarily placed there for their benefit or on behalf of any greater good.

Disinformation is proving to be an effective strategy for use in pursuit of political power or financial gain. During collective crises, like pandemics and environmental disasters, information becomes an issue of life and death. In a time of war, information is weaponized both for offense and defense. This was clearly demonstrated during Russia's invasion of Ukraine where the Internet provided a sociotechnical front for an accompanying information war.⁵

Corporations and governments have unprecedented access to private data at scale, and the people from whom it is taken have little control over how it is used. There is a pressing need to foster information literacy globally, a literacy that includes knowing how to critically assess the origin of information, becoming more skeptical of requests for private information, and understanding the risks that come with sharing it.

The quality of education in a community will either support or undermine the ability of its members to locate online information, ascertain its accuracy, and protect their privacy. In fact, education has emerged as one of the key factors driving information inequity

and the growing divide between the “information rich” and the “information poor.”

The digital divide is just one of many pressing information-related challenges. News headlines on any given day will offer stories pertaining to conspiracy theories and election interference, privacy and surveillance, disaster response and health disparities, structural inequalities, and algorithmic biases. These are all challenges that information science is uniquely positioned to address.

The practices of information science include the classification, organization, retrieval, analysis, utilization, governance, management, study, design, expression, and preservation of information. Information professionals may be librarians, records managers, analysts, programmers, researchers, administrators, technicians, designers, curators, or archivists, among other traditional, related, and evolving professions. Their titles may or may not include the term “information.” They may have been trained within any number of different intellectual traditions and degree programs, and make use of a wide array of tools, methods, practices, and technologies.

As scholars, information scientists conduct interdisciplinary research, both quantitative and qualitative, in lab and community settings, asking and seeking answers to questions across all domains of human knowledge and experience. In this introduction to the volume we begin by discussing the complexities of the term “information,” and the challenges of defining the field of information science. Then we lay out the organization of the book, and let readers know what they can expect to find here.

WHAT IS INFORMATION?

Information flows continuously around us, into us, and through us, and yet we may be unconscious of the role it plays in creating the worlds in which we live. Our bodies themselves are full of information; every moment of the day we take in and process sensory experiences that we sort and store in different parts of our brain depending on whether they are temporary working memories, episodic memories about things that have happened, or semantic memories relating to facts that we pick up about the world.

Procedural memories (the things we know how to do) are stored in both brain and body. Our DNA encodes genetic information, the instructions that guide our growth, development, and eventual decline. Human knowledge may one day be stored on synthesized DNA molecules, which can hold mind-boggling amounts of information in a space the size of the period at the end of this sentence.⁶

Everyone may agree that the phenomenon of study for the field of information science is *information*, but it is surprisingly difficult to define. The term is chimerical, with multiple academic definitions and commonsense usages. As a noun, it describes physical objects (books) or digital documents (files) that contain meaningful content encoded in text or images, but we also use the word to describe casual speech in which information is passed (gossip). We typically refer to both the container of information (an instruction sheet) and the meaning it conveys (the instructions) as information. The term sometimes serves as an adjective describing an attribute (an informational video) or a value (an informative meeting), but it also defines subjective states related to understanding (being well-informed) or status (being an informant). As a verb, information may be a relational act (informing someone else) or be implicated in complex social processes (the political, psychological, and social effects of dis/misinformation upon public opinion and behavior).

Information is sometimes conceptualized as a metaphorical living thing, which moves and grows, and which can be analyzed much as other living things are. It is situated within networks of relations, and changes through predictable life cycles. Information can be observed to have an identifiable moment of birth (a press release); from there it may change in unexpected ways (a conspiracy theory); it can be acted upon (an executive order), or suppressed (book banning), but sometimes it seems to have a life of its own (going viral). It has a life span (the news cycle), after which it may be forgotten (dementia-related memory loss), literally lost (phone dropped over a cliff), or destroyed (intentional, unrecoverable file deletion). Information management professionals use the metaphor of the information life cycle to organize the policies and practices involved with steering an organization's information resources through all their stages from creation to retirement (or destruction).

A more computational way of thinking about information comes from *information theory*, a branch of mathematics that models the processing, storage, and transmission of information. It understands the term “information” narrowly, as signals that can be distinguished from random noise. In this model, communication occurs when an encoded signal is transmitted to a receiver which successfully decodes it, despite it being corrupted by noise en route.⁷ Developed further by others, but first proposed by Claude Shannon, the model remains fundamental to modern communication systems. Because it treats information as a resolution of uncertainty, it is directly relevant to probability theory, pattern recognition, data compression, and cryptography. Further along this track is *quantum information science*, a theoretical and experimental field within physics that combines computational models with quantum mechanics.

Any attempt to define information comprehensively would have to contend not only with dramatically different academic understandings, but also commonsense notions that further muddy the water. Even if a definition were to be agreed upon and applied across disciplines, it would be unlikely to have much effect on everyday usage. For example, “information” is often presented as a discrete stage between “data” and “knowledge” on a path towards “wisdom.” This hierarchical progression is visualized in the data-information-knowledge-wisdom (DIKW) pyramid, which places data at the bottom and wisdom at the top.⁸ The *DIKW hierarchy* solidified a generally accepted idea that data is potential information, lacking meaning until analysis extracts meaningful information from it. Even so, the words “data” and “information” are often used as synonyms, even in scholarly writing. Likewise, knowledge is sometimes understood as emerging when information is applied to accomplish a practical task (knowledge about how to do something), but it is common to find the words “knowledge” used interchangeably with “information” as well.

The idea that knowledge is derived from information (rather than, say, the other way around) is disputed; since knowledge is a subjective experience, “knowable only to the knower,” only *information about* knowledge can be collected and shared.⁹ “Knowledge” and “wisdom” each have multiple context-bound cultural meanings

as well, further complicating the situation. The DIKW model as a whole has been criticized for assuming a direct hierarchical relationship between sociotechnical processes that are non-linear and entangled¹⁰ but it shows few signs of disappearing; it has been productively reinterpreted and applied within particular domains.¹¹

In the absence of consensus on how to define information, scholars face the ongoing work of classifying and mapping the definitions. Lyn Robinson and David Bawden productively analyze the “gap” between five distinct approaches to information: technological, physical, biological, social, and philosophical.¹² Marcia Bates suggests six broad approaches to defining information: semiotic, activity-based, propositional, structural, social, multi-type, and deconstructionist. Many information scientists and theorists from other disciplines have contributed to these various strains of thought.¹³

We are left to conclude that the contradictions attending the term information don’t seem to be particularly problematic for the field. Robinson and Bawden agree, concluding that there is insight to be found in attempts to bridge the gaps in understanding, even between dramatically different approaches. The practical consequence of the complexity, as Michael Buckland counsels, is that when it matters, we have to be precise.¹⁴ The rest of the time, contentious discussions around a key term can be productive and even on-point; consider how “society” and “culture” simultaneously bedevil and invigorate the fields of sociology and anthropology respectively.

The theoretical and practical work of information science is relevant across all domains of human understanding, from the artistic to the legal to the psychological to the astrophysical; each one understands information in relation to its domain. Far from being a liability, it is this extraordinary *breadth of relevance* that allows the interdisciplinary field of information science to respond to some of the world’s most vexing problems.

THE CHALLENGE OF DEFINING THE FIELD

In an ambitious study conducted from 2003 to 2005 an international panel of 57 scholars from 16 countries all weighed in on

the question. They ended up producing 50 definitions of information science.¹⁵ The problem (or the opportunity, depending on one's point of view), is that information science shares foundational histories, skills, methods, and practices not only with library science and other longstanding disciplines including archival science, history, computer science, communication, and statistics, among others, but also with more recent interdisciplinary fields such as informatics, science and technology studies (STS), media studies, and data science. Courses and degree programs with the word "information" in their name may exist within, above, or alongside any of these programs, or they may be housed idiosyncratically within newly devised interdisciplinary units.

The information technologies that emerged during the 20th century fell under the purview of multiple academic disciplines, triggering a sometimes contentious interplay between information theory as it was embraced by computer science and engineering, and the fields of library science and documentation, which became the discipline of library and information science (LIS) in the middle of the 20th century. Efforts to synthesize LIS with information theory largely foundered, leaving inquiry into the meaningful content, classification, and organization of information and information-related technologies on one side, and the structure and engineering of those technologies largely on the other.¹⁶

Academic boundary-making is an ongoing process; in the 21st century, programs offering information-related degrees vary greatly in their pedagogical missions. There is also significant international variation. In France for example, information science has been part of *information and communication sciences* since the 1970s, a circumstance that has challenged French information scientists to carve out an identity that is distinct both from computer science and from communication.¹⁷

The schools that teach information science may differ greatly, or hardly at all from one another. Some programs offer graduate degrees that include library certification and licensure, but some do not. Some include human-centered design practices and user experience research, but some do not; some focus on information organization and knowledge management as central concerns, but some do not; some view developments in data science which incorporate artificial intelligence and machine learning to be essential

areas of inquiry, but some do not; some include training in archival methods of information conservation and preservation, but some do not; an increasing number place diversity, equity, inclusion and critical approaches to data, information, and librarianship at the heart of their programs, but some do not. There are also schools that consider information science to be inclusive of all of these areas.

As a result, faculty employed in degree-granting information science programs tend to be methodologically diverse. They may or may not call themselves “information scientists.” They may retain multiple professional identities, making use of diverse theoretical and methodological approaches originating in STEM disciplines, the social sciences, the humanities, and the arts. They may have backgrounds in interdisciplinary fields such as informatics, which combines computational and information-related skills and practices with deeper study and application of those practices within a particular domain, or in science and technology studies (STS), in which historical, sociological, and theoretical frameworks shed light on sociotechnical processes. They may also be engaged in a transformation of traditional disciplines, creating new fields within which to study information.¹⁸

These unusually generative circumstances have the effect of shifting curricular conversations from the capacity (what are various programs on a particular campus able to teach) to the territory (which programs have the *right* to offer courses about which information-related topics, skills, and practices).¹⁹ So, the question of what information science *is* cannot be separated from the politics and economics of the academe. How the institutional title of Information Science(s) appears within any given institutional structure today will reflect national, regional, and local campus politics, and the pressing realities of limited resources.

Systems of higher education without long-term, sustainable funding models, are increasingly vulnerable to the shifting priorities of government and the marketplace. The relative prosperity of programs related to science and technology, compared to alarmingly defunded arts and humanities units, may be guiding the strategic naming of programs, rather than substantive discussions about how, in some more ideal situation, an interdisciplinary program should be organized, institutionally situated, and governed.

It is often suggested that, because of its heterogeneity, information science should forego the label of “discipline” altogether, and simply refer to itself as a “field of study,” a term that designates a broad intellectual area closely associated with the skills and practices of the professions that compose it.²⁰ Some degree-granting programs organize themselves around the mission of training students to take up positions in information-related professions, referring to the field as “Information Studies.” The truncation of “Library and Information Science” down to “Information Science” sometimes goes one step further to refer to the field as “Information.” Another strategy is to cast information science into the plural form, “information sciences,” signaling wider inclusivity of interest and practice.

Because of the breadth of application of information science to all domains of human inquiry, enterprise, and creativity, it has also been described as a *metadiscipline*, much like education, which gives its pedagogical consideration to every academic discipline, including its own. The institutional solidity and longevity of education provide information science with the assurance that as a discipline it can be capacious while remaining a coherent intellectual enterprise.²¹

The question of what information science is, and where it belongs in the greater academic community, has been under discussion for the last half-century within national and international professional organizations, including the Association for Library and Information Science Education (ALISE) and the Association for Information Science and Technology (ASIS&T).

The iSchool Movement was sparked by conversations during a 1988 ALISE meeting which proposed centering the field on the needs of the information professions. It began as a consortium of academic institutions, with a founding “Gang of Three” (University of Pittsburgh, Syracuse University, and Drexel University), expanding into gangs of increasing numbers until it became a caucus. Over 100 institutions across the globe now refer to themselves as iSchools.

The consensus opinion of the iSchool Caucus is that humanities, social science, and STEM approaches are all necessary aspects of academic training for information professionals. This is typically expressed through value statements referring to the relationships

between “information, technology, and people.” While the official iSchool website refers to the field as “information,” individual iSchools go their own ways in naming themselves. The iSchool movement is just one way that information science is evolving,²² and even among iSchools there is significant variation in the specific courses of study, professional licensures, and academic degrees offered.

The authors of this book are faculty in a college at the University of Illinois Urbana-Champaign that was long known as the Graduate School of Library and Information Science (GSLS). After it joined the iSchool caucus it renamed itself the “School of Information Sciences.” It remains a global leader in library science graduate education, deeply involved in the intellectual communities of LIS. The name change, which did not occur without contention, reflects the argument that the field of information science must increase its reach if it is to respond effectively to rapidly evolving information systems, technologies, and sociotechnical practices, and to provide comprehensive training for all information professions: the traditional, the emerging, and the yet-to-be-imagined.

AN INVITATION TO INFORMATION SCIENCE

Sociologist C. Wright Mills described the “sociological imagination” as a lens through which individuals can view their individual lives within broader social contexts. During the social upheavals of the 1960s, sociology issued an invitation to students, and to the public at large, to try looking through that lens. In his book *Invitation to Sociology* Peter Berger described the unique methodologies of sociology, which were trained on social factors rather than on individual metrics, as a vantage from which to understand the socially constructed nature of the world and its persistent stratifications. Generations of social theorists across many disciplines who responded to that call have had a profound effect on intellectual conversations still ongoing today.²³

The lens of the sociological imagination is as relevant now as it ever was. As continuously evolving information technologies reshape our social practices and norms, there is a critical need for a shared consciousness of the ways in which information practices, processes, systems, and technologies are shaped by, and shape, our

shared social worlds. It is commonplace to say that we are in an *Information Age*. For some, this age is defined by the dramatic transformations in information and communication technologies that occurred over the last century, in the leap from analog to digital storage, which produced the Internet, cellular phones, and then faster (and smaller and stranger) methods of information sharing, storage, search, and retrieval. But for others, what marks this moment are the social changes that have driven those technologies, and the consequences that have resulted from them.

We have more information of every kind at our (literal) fingertips than we can digest. It has become less an insight than a truism that we can feel that we are drowning in information. And there are other dangers as well; the technologies that enable crucial flows of information to meet contemporary crises, have also produced vulnerabilities that enable bad actors to take advantage of the 21st-century information cornucopia heralded as “Big Data.”

We have arrived at another game-changing intellectual moment, in which students and scholars are being called to study the swiftly evolving information challenges of our time. In crucial capacities, the interdisciplinary agility of information science can well serve. By making information and its processes visible, information professionals can refine the tools, technologies, and networks through which information is collected and organized, is impeded or flows, and is accessed within and between systems, domains, organizations, and nations. At the same time, information science works to mitigate biases within information systems, identifying inequities in access to information, in who benefits from new technologies, and in who pays the social and environmental costs.

We accept the view, shared by many, that information science is best described as an interdisciplinary field of study, closely associated with, and reflecting the needs of, its professions. But we are also persuaded that information science is more than its practices, professions, tools, and technologies, and that it may also be productively understood as a metadiscipline, with its unique lens trained on every domain of human inquiry, including its own.

In this book, we invite readers to enter a complex intellectual terrain. All maps are partial; we acknowledge that others, situated differently, would chart a different route and highlight different

points of interest along the way. Information science is here understood to have emerged from the field of library and information science, enlivened by information-related technologies, skills, and practices shared with many other disciplines and interdisciplinary fields. We take an inclusive approach which situates the study of information at the conceptual, virtual, and material places where information, technology, and society meet, embracing multiple methodological, theoretical, and domain-specific approaches to understanding what takes place there. We refer to *information science* in the singular when referring to the field, to its expression within information professions, and to information-related concepts and practices. In thinking about its institutional expressions, we recognize that the complexity and contradictions of its manifestations make it rest more comfortably within the inclusive plural: *Information Sciences*.

Narratively, we trace the path of information across its metaphorical life cycle. The body of this book follows the life of information as it is collected (Chapter 1), classified (Chapter 2), organized (Chapter 3), retrieved (Chapter 4), utilized (Chapter 5), governed (Chapter 6), studied (Chapter 7), designed (Chapter 8), and preserved (Chapter 9). In the book's concluding discussion, we address imagined futures for information and information science (Chapter 10). Throughout the book we highlight the contributions that the field can make toward addressing the serious information challenges of our day, and present critical perspectives on the ideas and practices described.

Information Science: The Basics invites a new generation to the work of making visible (and legible to all) the contexts through which information flows powerfully and continuously through the backgrounds, and the foregrounds, of our lives.

NOTES

- 1 Jesse Gary, "Photo of Girls Using Taco Bell Wifi become Symbol of Digital Divide," 31 August 2020, KTVU FOX, 2. <https://www.ktvu.com/news/photo-of-girls-using-taco-bell-wifi-becomes-symbol-of-digital-divide>.
- 2 Kevin de León, @kdeleon, *Twitter*, 28 August 2020. <https://pic.twitter.com/ijPXvcxsLQ>.

- 3 Kevin de León. <https://twitter.com/kdeleon/status/1299386969873461248>. Retrieved 12 May 2022.
- 4 “Monumental Broadband Legislation by Senator Lena Gonzalez and Assemblymember Cecilia Aguiar-Curry Signed Into Law,” 8 October 2021. <https://sd33.senate.ca.gov/news/2021-10-08-monumental-broadband-legislation-senator-lena-gonzalez-and-assemblymember-cecilia>.
- 5 Jim Allsop, “The Ongoing Information War over Ukraine,” *The Media Today*, *Columbia Journalism Review*, 9 May 2022.
- 6 Ed Yong, “This Speck of DNA Contains a Movie, a Computer Virus, and an Amazon Gift Card,” *The Atlantic*, 2 March 2017.
- 7 William F. Aspray, “The Scientific Conceptualization of Information: A Survey,” *IEEE Annals of the History of Computing* 7, no. 2 (1985): 117–140.
- 8 Russell L. Ackoff, “From Data to Wisdom,” *Journal of Applied Systems Analysis* 16, no. 1 (1989): 3–9.
- 9 Thomas D. Wilson, “Human Information Behavior,” *Informing Science* 3, no. 2 (2000): 50.
- 10 David Weinberger, “The Problem with the Data-Information-Knowledge-Wisdom Hierarchy,” *Harvard Business Review* 2 (2010). <https://hbr.org/2010/02/data-is-to-info-as-info-is-not>.
- 11 See for example Kate McDowell, “Storytelling Wisdom: Story, Information, and DIKW,” *Journal of the Association for Information Science and Technology* 72, no. 10 (2021): 1223–1233.
- 12 Lyn Robinson and David Bawden, “Mind the Gap: Transitions between Concepts of Information in Varied Domains,” in *Theories of Information, Communication and Knowledge* (Dordrecht: Springer, 2014): 121–141.
- 13 Marcia J. Bates, “Information,” *Encyclopedia of Library and Information Sciences* 3 (2010): 2048–2063.
- 14 Michael Buckland, *Information and Society* (Cambridge, MA: MIT Press, 2017): 4.
- 15 Chaim Zins, “Conceptions of Information Science,” *Journal of the American Society for Information Science and Technology* 58, no. 3 (2007): 335–350.
- 16 Ronald R. Kline. “What is Information Theory a Theory of? Boundary Work among Information Theorists and Information Scientists in the United States and Britain during the Cold War,” in *The History and Heritage of Scientific and Technological Information Systems*, eds. W. B. Rayward and M. E. Bowden (Medford, NJ: American Society of Information Science and Technology and the Chemical Heritage Foundation, 2004): 15–28.
- 17 Fidelia Ibekwe-Sanjuan, “Whither Information Science in France?” in *International Perspectives on the History of Information Science and Technology* (Baltimore, MD: Association of the American Society for Information Science and Technology, 2012): 83–95.
- 18 See Luciano Floridi, *The Philosophy of Information* (Oxford: OUP, 2013).
- 19 See Lili Luo, “Being Interdisciplinary: A Look into the Background and Experiences of iSchool Faculty Members,” *LIBRES: Library and Information Science Research Electronic Journal* 23, no. 2 (2013): 1–20.

- 20 David Bawden and Lyn Robinson. *Introduction to Information Science* (London: Facet Publishing, 2015): 3; Charles H. Davis, and Debora Shaw, eds. *Introduction to Information Science and Technology* (Medford, NJ: American Society for Information Science and Technology, 2011): 16.
- 21 Marcia J. Bates, "The Information Professions: Knowledge, Memory, Heritage." *Information Research* 20, no. 1 (2015). Retrieved from <http://InformationR.net/ir/20-1/paper655.html>.
- 22 See Joacim Hansson, *Educating Librarians in the Contemporary University: An Essay on iSchools and Emancipatory Resilience in Library and Information Science* (Sacramento, CA: Library Juice Press, 2019).
- 23 Peter Berger, *Invitation to Sociology* (New York, NY: Anchor Books, 1963).

References

- Abend, Lisa. . "Meet the Lithuanian 'Elves' Fighting Russian Disinformation." Time. 6 March 2022. <https://time.com/6155060/lithuania-russia-fighting-disinformation-ukraine/>.
- Abid, Abubakar. , Maheen. Farooqi , and James. Zou . "Large Language Models Associate Muslims with Violence." Nature Machine Intelligence 3, no. 6 (2021): 461–463.
- "About." OpenAI. Retrieved 17 June 2022. <https://openai.com/about/>.
- "About." The Information Behavior Conference. <https://pages.cms.hu-berlin.de/ibi/isic-2022/about/>. Retrieved 8 June 2022.
- "About DCMI." Dublin Core Metadata Initiative Website. Retrieved 20 May 2020. <https://www.dublincore.org/about/>.
- "About the Internet Archive." Internet Archive. <https://archive.org/about/>. Retrieved 24 May 2022.
- "About Us." Europeana. <https://www.europeana.eu/en/about-us>. Retrieved 11 June 2022.
- "About Us." United States Indigenous Data Sovereignty Network. <https://usindigenousdata.org/about-us>. Retrieved 22 June 2022.
- Ackoff, Russell L. "From Data to Wisdom." Journal of Applied Systems Analysis 16, no. 1 (1989): 3–9.
- Adams, Douglas. , The Hitchhiker's Guide to the Galaxy. Pan Books, 1979.
- "Adelina the Jester." Open Domesday. <https://opendomesday.org/name/adelina-the-jester/>. Retrieved 30 May 2022.
- Adu, Kofi Koranteng. , and Patrick. Ngulube . "Key Threats and Challenges to the Preservation of Digital Records of Public Institutions in Ghana." Information, Communication & Society 20, no. 8 (2017): 1127–1145.
- Ahmed, Syed Masud. , Göran. Tomson , Max. Petzold , and Zarina Nahar. Kabir . "Socioeconomic Status Overrides Age and Gender in Determining Health-Seeking Behaviour in Rural Bangladesh." Bulletin of the World Health Organization 83 (2005): 109–117.
- "AI Global Surveillance Technology." Carnegie Endowment for International Peace. <https://carnegieendowment.org/publications/interactive/al-surveillance>. Retrieved 4 June 2022.
- Albert-László, Barabási. . "Network Science." Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences 371. 1987 (2013): 20120375.
- Allen, Walter C. , and Robert F. Delzell . Ideals and Standards: The History of the University of Illinois Graduate School of Library and Information Science, 1893–1993. The Graduate School of Library and Information Science, University of Illinois at Urbana-Champaign, 1992.
- Allsop, Jim. . "The Ongoing Information War over Ukraine." The Media Today, Columbia Journalism Review, 9 May 2022.
- American Library Association . "ALA Statement on Censorship of Information Addressing Racial Injustice, Black American History, and Diversity Education." Retrieved 17 May 2022. <https://www.ala.org/advocacy/intfreedom/statement/opposition-censorship-racial-injustice-black-history-diversity-education>.

"Ancient Mesopotamia." *Journal of Technical Writing and Communication* 49, no. 3 (2019): 338–364.

Anderson, Theresa Dirndorfer. , and Jo. Orsatti . "'Rhythms of Being' at ISIC- Understanding the Place of the ISIC Conferences in Information Seeking Research." *Information Research* 13 no. 4 (2008). <http://informationr.net/ir/13-4/paper381.html>. Retrieved 22 September 2022.

"The Anglo-Saxon Chronicle: Eleventh Century." The Avalon Project: Documents in Law, History & Diplomacy. Yale Law School. <https://avalon.law.yale.edu/medieval/ang11.asp>. Retrieved 30 May 2022.

Apenīte, Māriņe. . "Subject Indexing at the National Library of Latvia: New Approach, Challenges, and Benefits." *Cataloging & Classification Quarterly* 59, no. 4 (2021): 334–354.

Appelbaum, Ralph. . "Activating Public Space." In *Information Architects*, edited by Richard Saul Wurman and Peter. Bradford , 150–161. Zurich: Graphis Inc., 1996.

Archi, Alfonso. , "Ebla and Its Archives." In *Texts, History, and Society*, De Gruyter, 2015.

Arif, Ahmer. , Leo Graiden. Stewart , and Kate. Starbird . "Acting the Part: Examining Information Operations within #BlackLivesMatter Discourse." *Proceedings of the ACM on Human-Computer Interaction* 2, no. CSCW (2018): 1–27.

"ARLIS/NA Statement on Removal of Library of Congress: Illegal Aliens – Subject Headings." *Art Libraries Association of North America*. 21 January 2022.

Arora, Payal. . "Bottom of the Data Pyramid: Big Data and the Global South." *International Journal of Communication* 10 (2016): 19.

Ash, Thomas. . "Naudé, Mazarin and the Origins of France's Oldest Public Library." March 9, 2017. <https://thomasash.wordpress.com/2017/03/09/naude-mazarin-and-the-origins-of-frances-oldest-public-library/>.

Aspray, William F. "The Scientific Conceptualization of Information: A Survey." *IEEE Annals of the History of Computing* 7, no. 02 (1985): 117–140.

Assari, Shervin. , Cleopatra H. Caldwell , and Mohsen. Bazargan . "Association between Parental Educational Attainment and Youth Outcomes and Role of Race/Ethnicity." *JAMA Network Open* 2, no. 11 (2019). <https://doi.org/10.1001/jamanetworkopen.2019.16018>

Astuti, Santi Indra. , Lumakto. Giri , and Nuril. Hidayah . "Video Web Drama Series for Combating Disinformation." *Aspiration Journal* 1, no. 1 (2020): 7.

Audunson, Ragnar. , Herbjørn. Andresen , Cicilie. Fagerlid , Erik. Henningsen , Hans-Christoph. Hobohm , Henrik. Jochumsen , Håkon. Larsen , and Tonje. Vold . *Libraries, Archives and Museums as Democratic Spaces in a Digital Age*. Berlin: De Gruyter, 2020.

Ayad, Moustafa. . "The Vladimirror Network: Pro-Putin Power-Users on Facebook." *Institute of Strategic Dialogue*. 4 April 2022. <https://www.isdglobal.org/isd-publications/the-vladimirror-network-pro-putin-power-users-on-facebook/>.

"The Bad News Game." *DebunkEU*. <https://www.getbadnews.com/en/play>. Retrieved 9 June 2022.

Baffour-Awuah, Margaret. . "The Carnegie Revitalisation of Public Libraries in Africa: A Possible Tool for Breaking Down Barriers to School Library Development." In IASL Annual Conference Proceedings, 2021. <https://doi.org/10.29173/iasl7545>

Banks, R.R. and Ford, R.T. "(How) Does Unconscious Bias Matter: Law, Politics, and Racial Inequality." *Emory LJ* 58 (2008).

Bannon, Liam. , Jeffrey. Bardzell , and Susanne. Bødker . "Introduction: Reimagining Participatory Design: Emerging Voices." *ACM Transactions on Computer-Human Interaction (TOCHI)* 25, no. 1 (2018): 1–8.

Baron, Jill E. , and Tina. Gross . "Sorrow, Fury, Helplessness, and Cynicism: An Account of the Library of Congress Subject Heading 'Illegal Aliens.'" In *Borders and Belonging: Critical Examinations of Library Approaches toward Immigrants*, edited by Ana. Ndumu . Sacramento, CA: Library Juice Press, 2021.

Basile, Jonathan. . *Tar for Mortar: "The Library of Babel" and the Dream of Totality*. Sacramento, CA: Punctum Books, 2018.

Bates, Marcia J. "Information." *Encyclopedia of Library and Information Sciences* 3 (2010): 2048–2063.

Bates, Marcia J. "Information Behavior." In *Encyclopedia of Library and Information Sciences*, edited by John D. McDonald and Michael. Levine-Clark . Boca Raton: CRC Press, 2017: 2074–2085.

Bates, Marcia J. "The Information Professions: Knowledge, Memory, Heritage." *Information Research* 20, no. 1 (2015). Retrieved from <http://InformationR.net/ir/20-1/paper655.html>

Batool, Syeda Hina. , and Sheila. Webber . "A Contextual Framework for Primary Education: Fostering Information Literacy in Pakistan." *Global Knowledge, Memory and Communication* 68, no. 3 (2018): 164–176.

Battle-Baptiste, Whitney. , and Britt. Rusert , editors. *WEB Du Bois's Data Portraits: Visualizing Black America*. San Francisco, CA: Chronicle Books, 2018.

Bawden, David. . "Users, User Studies and Human Information Behaviour: A Three-Decade Perspective on Tom Wilson's 'On User Studies and Information Needs.'" *Journal of Documentation* 62, no. 6 (2006): 671–679.

Bawden, David. , and Lyn. Robinson . *Introduction to Information Science*. London: Facet Publishing, 2015.

Becker, Annette. . "Memory Gaps: Maurice Halbwachs, Memory and the Great War." *Journal of European studies* 35, no. 1 (2005): 102–113.

Beel, David E. , Claire D. Wallace , Gemma. Webster , Hai. Nguyen , Elizabeth. Tait , Marsalli. Macleod , and Chris. Mellish . "Cultural Resilience: The Production of Rural Community Heritage, Digital Archives and the Role of Volunteers." *Journal of Rural Studies* 54 (2017): 459–468.

Belghith, Yasmine. , Sukrit. Venkatagiri , and Kurt. Luther . "Compete, Collaborate, Investigate: Exploring the Social Structures of Open Source Intelligence Investigations." In *CHI Conference on Human Factors in Computing Systems*, 2022: 1–18. <https://doi.org/10.1145/3491102.3517526> .

Belitz, Clara. , Jaclyn. Ocumpaugh , Steven. Ritter , Ryan S. Baker , Stephen E. Fancsali , and Nigel. Bosch . "Constructing Categories: Moving beyond Protected Classes in Algorithmic Fairness." *Journal of the Association for Information Science and Technology* (2022).

<https://doi.org/10.1002/asi.24643>.

Bender, Emily M. "Human-Like Programs Abuse Our Empathy – Even Google Engineers Aren't Immune." *The Guardian*, 14 June 2022.

Berman, Sanford. *Prejudices and Antipathies: A Tract on the LC Subject Heads Concerning People*. Metuchen, NJ: Scarecrow Press, 1971.

Bernal, J. D. "The Transmission of Scientific Information: A User's Analysis." In *Proceedings of the International Conference on Scientific Information* 958 (1958): 77–95. Cited in Wilson, Thomas D. "Human Information Behavior." *Informing Science* 3, no. 2 (2000): 50.

Berners-Lee, T. , Hendler, J. , & Lassila, O. "The Semantic Web." *Scientific American*, 284(5), (2001). 28–37.

Bertot, John Carlo. , Paul T. Jaeger , Lesley A. Langa , and Charles R. McClure . "Public Access Computing and Internet access in Public Libraries: The Role of Public Libraries in E-Government and Emergency Situations." *First Monday* (2006).

Beyer, Wiebke. . "Teaching in Old Babylonian Nippur, Learning in Old Assyrian Aššur?" *Studies in Manuscript Cultures*, 15: (2021).

Bhaskar, Chakravorti. . "Why AI Failed to Live Up to Its Potential during the Pandemic." *Harvard Business Review*. 27 March 2022.

Bibliothèque nationale de France . <https://gallica.bnf.fr/>. Retrieved 23 May 2022.

Bizer, Christian. , Tom. Heath , and Tim. Berners-Lee . "Linked Data: The Story So Far." In *Semantic Services, Interoperability and Web Applications: Emerging Concepts*, edited by Amit P. Sheth . Hershey, PA: IGI Global, 2011: 205–227.

Blair, Ann M. *Too Much to Know*. New Haven, CT: Yale University Press, 2010.

Blake, Virgil LP. . "Forging the Anglo-American Cataloging Alliance: Descriptive Cataloging, 1830–1908." *Cataloging & Classification Quarterly* 35, no. 1–2 (2002): 3–22.

Blei, Daniela. . "How the Index Card Cataloged the World." *The Atlantic*. 1 December 2017.

Bødker, S. , & Kyng, M. *Participatory Design That Matters: Facing the Big Issues*. *ACM Transactions on Computer-Human Interaction* 25, no. 1 (2018): 12.

Bogaert, Hannah. . "History Repeating Itself: The Resurgence of the Taliban and the Abandonment of Afghan Women." *Immigration and Human Rights Law Review* 4, no. 1 (2022): 3.

Bohannon J. and A. Elbakyan "Data from: Who's Downloading Pirated Papers? Everyone." *Dryad Digital Repository* (2016).

<https://doi.org/10.5061/dryad.q447c>

Bond, Sarah E. "Erasing the Face of History." *New York Times*. 14 May 2011.

Bonn, Maria. . "Computation, Corpus, Community: The Hathitrust Research Center Today." *College & Research Libraries News* 77, no. 4 (2016): 194–197.

Bonn, Maria. , Lori. Kendall , and Jerome. McDonough . "Libraries and Archives and the Preservation of Intangible Cultural Heritage: Defining a

Research Agenda" (white paper). Champaign, IL: School of Information Sciences (2017): 50–51. Retrieved 22 September 2022. <http://hdl.handle.net/2142/97228>.

"The Book of Books: Hernando Colón's Libro de los Epítomes." Department of Nordic Studies and Linguistics, University of Copenhagen. Retrieved 17 May 2022. <https://nors.ku.dk/english/research/arnamagnaeon/the-book-of-books/>.

Borges, Jorge Luis. . "The Analytical Language of John Wilkins." Other Inquisitions. 1952 (1937): 103.

<https://ccrma.stanford.edu/courses/155/assignment/ex1/Borges.pdf>.

Borges, Jorge Luis. . "On Exactitude in Science." In *Collected Fictions*, edited by Jorge Luis. Borges , translated by Andrew. Hurley . New York: Penguin Putnam, 1998.

Borges, Jorge Luis. . "The Total Library." In *Selected Non-Fictions*, edited by Esther. Allen , Suzanne Jill. Levine , and Weinberger and translated by Eliot. Weinberger . New York: Viking, 1999.

Borges, Jorge Luis. . *The Library of Babel*. Translated by Andrew. Hurley . Boston, MA: David R Godine Press, 2000.

Borges, Jorge Luis. . *Labyrinths: Selected Stories & Other Writings*. Vol. 1066. New York, NY: New Directions Publishing, 2007.

Brasseur, Lee. . "Florence Nightingale's Visual Rhetoric in the Rose Diagrams." *Technical Communication Quarterly* 14, no. 2 (2005).

Breen, Richard. , and Walter. Müller . *Education and Intergenerational Social Mobility in Europe and the United States*. Stanford CA: Stanford University Press, 2020.

Brewer, Graham Lee. . "Lakota Elders Helped a White Man Preserve Their Language. Then He Tried to Sell It Back to Them." *NBC News*, 3 June 2022.

"A Brief History of Crowdsourcing [Infographic]." *Crowdsourcing.org*. 18 March 2012. Archived from the original on 3 July 2015.

Briet, Suzanne. , and Laurent. Martinet . *What Is Documentation?: English Translation of the Classic French Text*. Lanham, MD: Scarecrow Press, 2006.

Browne, Simone. . "Dark Matters." In *Dark Matters*. Durham, NC: Duke University Press, 2015: 104.

Brumm, Adam, Adhi. Agus Oktaviana , Basran. Burhan , Budianto. Hakim , Rustan. Lebe , Jian-xin. Zhao , Priyatno Hadi. Sulistyarto et al. "Oldest Cave Art Found in Sulawesi." *Science Advances* 7, no. 3 (2021).

<https://doi.org/10.1126/sciadv.abd4648>

Buckland, Michael K. "Information as Thing." *Journal of the American Society for information science* 42, no. 5 (1991a): 351–360; and Buckland, Information Society, p. 22.

Buckland, Michael K. "Information Retrieval of More Than Text." *Journal of the American Society for Information Science* 42, no. 8 (1991b): 586–588.

Buckland, Michael K. "A Brief Biography of Suzanne Renée Briet." *Science* 42, no. 5 (1991b): 351–360.

Buckland, Michael K. "Emanuel Goldberg, Electronic Document Retrieval, and Vannevar Bush's Memex." *Journal of the American Society for Information Science* 43, no. 4 (1992): 284–294.

Buckland, Michael K. "What Is a 'Document'?" *Journal of the American Society for Information Science* 48, no. 9 (1997): 805.

Buckland, Michael K. "Histories, Heritages, and the Past: The Case of Emanuel Goldberg." In *The History and Heritage of Scientific and Technical Information Systems*, edited by W. B. Rayward and M. E. Bowden. Medford, NJ: Information Today, 2004: 39–45.

Buckland, Michael. *Information and Society*. Cambridge, MA: MIT Press, 2017.

Burke, Heather. , and Claire. Smith. "Timeline." In *Kennewick: Perspectives on the Ancient One*, edited by Heather. Burke , Dorothy. Lippert , Claire. Smith , Joe. Watkins , and Larry. Zimmerman . Walnut Creek, CA: Left Coast Press, 2008.

Burrage, Rachel L. , Sandra L. Momper , and Joseph P. Gone . "Beyond Trauma: Decolonizing Understandings of Loss and Healing in the Indian Residential School System of Canada." *Journal of Social Issues* 78, no. 1 (2022): 27–52.

Burt, Laura. . "Vivian Harsh, Adult Education, and the Library's Role as Community Center." *Libraries & the Cultural Record* 44, no. 2 (2009): 234–255.

Bush, Vannevar. . "As We May Think." *The Atlantic Monthly* 176, no. 1 (1945): 101–108.

Bush, Vannevar. . "We Are in Danger of Building a Tower of Babel." *Public Health Reports* 68, no. 2 (1953): 149.

Busse, Timothy J. "Crossing the Digital Rubicon: Google Books and the Dawn of an Electronic Literature Revolution." *Houston Business and Tax Law Journal* 18 (2018): 121–149.

Cadwalladr, Carol. . "Are the Robots About to Rise? Google's New Director of Engineering Thinks So." *The Guardian*, 22 February 2014.

Card, Stuart K. , Thomas P. Moran , and Allen. Newell . *The Psychology of Human-Computer Interaction*. Boca Raton, FL: CRC Press, 1983.

Carroll, Lewis. . *Sylvie and Bruno Concluded*. London: Macmillan, 1894.

Cassino, Daniel. , and Peter. Wooley . "Some News Leaves People Knowing Less." *Farleigh Dickinson University Public Mind Poll*, November 21, 2011.

Cassoli, Federico. , Giulia. Fronda , and Michela. Balconi . "Human–Co-Bot Interaction and Neuroergonomics: Co-Botic vs. Robotic Systems." *Frontiers in Robotics and AI* 8 (2021): 659319.

Casson, Lionel. . *Libraries in the Ancient World*. New Haven, CT: Yale University Press, 2001.

Cat Zakrzewski . "Facebook Whistleblower Alleges Executives Misled Investors about Climate, Covid Hoaxes in New SEC Complaint." *The Washington Post*, 18 February 2022.

Cataloging Policy and Support Office . "Library of Congress Subject Headings Pre- vs. Post-Coordination and Related Issues: Report for Beacher Wiggins, Director, Acquisitions & Bibliographic Access Directorate." *Library Services, Library of Congress*, 2007: 1.

Cernat, Vasile. . "Roma Undercount and the Issue of Undeclared Ethnicity in the 2011 Romanian Census." *International Journal of Social Research Methodology* 24, no. 6 (2021): 761–766.

Chan, Anita. . "Big Data Interfaces and the Problem of Inclusion." *Media, Culture & Society* 37, no. 7 (2015): 1078–1083.

Chandler, Chris. , host, "Where Did IA Go? With Donna Spencer." *UX Radio* (podcast), 4 March 2021. <http://ux-radio.com/2021/03/where-did-ia-go-with-donna-spencer/>.

Charr, Manuel. . "How Are Museums Using Chatbots?" *Museum Next*, 14 September 2019.

Chen, Can. , and Sukumar. Ganapati . "Do Transparency Mechanisms Reduce Government Corruption? A Meta-Analysis." *International Review of Administrative Sciences* (2021): 00208523211033236.

Chen, Sibo. . "The Materialist Circuits and the Quest for Environmental Justice in ICT's Global Expansion." *tripleC: Communication, Capitalism & Critique. Open Access Journal for a Global Sustainable Information Society* 14, no. 1 (2016): 121–131.

"Christopher Columbus' Son Had an Enormous Library. Its Catalog Was Just Found." *All Things Considered*. National Public Radio. 24 April 2019.

Chou, Rose L. , Annie. Pho , and Charlotte. Roh . *Pushing the Margins: Women of Color and Intersectionality in LIS*. Sacramento, CA: Library Juice Press, 2018.

Clark, Shawn. and Ruth. Wylie . "Surviving a Cultural Genocide: Perspectives of Indigenous Elders on the Transfer of Traditional Values." *Journal of Ethnic and Cultural Studies* 8, no. 2 (2021): 316–346.

Colón-Aguirre, Mónica. , and Janet Ceja. Alcalá . "Everyday Information Practices of Migrant Latinas Living in Boston." In *Social Justice Design and Implementation in Library and Information Science*, edited by Bharat Mehra. London: Routledge, 2021, 116–132.

Condon, Margarte M. , and Elizabeth M. Hallam . "Government Printing of the Public Records in the Eighteenth Century." *Journal of the Society of Archivists* 7, no. 6 (1984): 348–388.

Connolly, Nuala. , and Claire. McGuinness . "Towards Digital Literacy for the Active Participation and Engagement of Young People in a Digital World." *Young People in a Digitalised World* 4 (2018): 77.

"Constitution of the Republic of Ecuador." *Political Database of the Americas*. Retrieved 19 May 2022.
<https://pdba.georgetown.edu/Constitutions/Ecuador/english08.html>.

Cook, Edward Tyas. . *The Life of Florence Nightingale*, Vol. 1. London: Macmillan, 1913.

Cooke, Nicole A. , Renate. Chancellor , Yasmeen. Shorish , Sarah Park. Dahlen , and Amelia. Gibson . "Once More for Those in the Back: Libraries Are Not Neutral." *Publishers Weekly*. 10 June 2022;

Cox, Andrew M. "Learning Bodies: Sensory Experience in the Information Commons." *Library & Information Science Research* 41, no. 1 (2019): 58–66.

Culliford, Elizabeth . "Rohingya Refugees Sue Facebook for \$150 Billion over Myanmar Violence." *Reuters*, 8 December 2021.

Cutter, Charles Ammi. . "The Buffalo Public Library in 1883." In *Papers and Proceedings of the Sixth General Meeting of the American Library Association*, Held at Buffalo, August 14 to 17, 1883. Press of Rockwell and Churchill, 1883, 49–55.

Cutter, Charles Ammi. . Expansive Classification: Part I: The First Six Classifications. C. A. Cutter, 1891–93.

Cy. Jillian Weise . “How a Cyborg Challenges Reality.” The New York Times, 22 June 2022. <https://www.nytimes.com/2022/06/22/special-series/cyborgs-reality-identity.html>. Retrieved 22 September 2022.

Dali, Keren. . “The Lifeways We Avoid: The Role of Information Avoidance in Discrimination against People with Disabilities.” *Journal of Documentation* 74, no. 6 (2018): 1258–1273.

“Dali Lives.” Retrieved 18 June 2022. <https://thedali.org/exhibit/dali-lives/>.

Darch, Peter T. “Managing the Public to Manage Data: Citizen Science and Astronomy.” arXiv preprint arXiv:1703.00037 (2017): 27.

“Darwin Core.” Taxonomic Databases Working Group. Retrieved 21 May 2022. <https://www.tdwg.org/community/dwc/>.

Davis, Charles H. , and Debora. Shaw , editors. *Introduction to Information Science and Technology*. Medford, NJ: American Society for Information Science and Technology, 2011.

de León, Kevin. (@kdeleon). Twitter. 28 August 2020a. <https://twitter.com/kdeleon/status/1299386969873461248>.

de León, Kevin. (@kdeleon). Twitter. 28 August 2020b. <https://pic.twitter.com/iJPXvcxsLQ>.

De Solla Price, Derek J. “An Ancient Greek Computer.” *Scientific American* 200, no. 6 (1959): 60–67.

De Solla Price, Derek. . “Gears from the Greeks.” *Journal for the History of Astronomy* 8 (1977): 143.

Deene, Joris. . “The Influence of the Statute of Anne on Belgian Copyright Law.” In *Global Copyright*. Cheltenham: Edward Elgar Publishing, 2010.

Delbert, Caroline. . “Beethoven Never Finished His Last Symphony. Can Robots Complete the Job?” *Popular Mechanics*, 18 December 2019.

Diesner, Jana. . “Small Decisions with Big Impact on Data Analytics.” *Big Data & Society* 2, no. 2 (2015): 2053951715617185.

“Digital Einstein Experience.” Retrieved 18 June 2022. <https://einstein.digitalhumans.com/>.

“Dive into Intangible Cultural Heritage!” UNESCO Intangible Cultural Heritage. Retrieved 11 June 2022. <https://ich.unesco.org/en/dive>.

“The Doctrine of Discovery, 1943.” The Gilder Lehrman Institute of American History. Retrieved 17 May 2022. <https://www.gilderlehrman.org/history-resources/spotlight-primary-source/doctrine-discovery-1493>.

Dolata, Mateusz. , and Kiram. Ben Aleya . “Morphological Analysis for Design Science Research: The Case of Human-Drone Collaboration in Emergencies.” In *International Conference on Design Science Research in Information Systems and Technology*, edited by Andreas. Drechsler , Aurlon Gerber. , Alan Hevner. . Springer, 2022, 17–29.

Dong, XinQi. , E-Shien. Chang , Melissa. Simon , and Esther. Wong . “Sustaining Community-University Partnerships: Lessons Learned from a Participatory Research Project with Elderly Chinese.” *Gateways: International Journal of Community Research and Engagement* 4 (2011): 31–47.

Dossey, Barbara Montgomery. . *Florence Nightingale: Mystic, Visionary, Reformer*. Springhouse, PA: Lippincott, Williams, and Wilkins, 2000: 74. Cited by Brasseur, "Florence," 162.

Douglas, Jennifer. . "Origins and Beyond: The Ongoing Evolution of Archival Ideas about Provenance." *Currents of Archival Thinking* 2 (2017): 25–52.

Dreazen, Yochi. . "The Brazen Bibliophiles of Timbuktu: How a Sneaky Team of Librarians Duped Al Qaeda." *The New Republic*, 2013.

Duchesneau, Michel. . "Gallica: The Online Digital Library of the Bibliothèque nationale de France." *Nineteenth-Century Music Review* 11, no. 2 (2014): 337–347.

Dzhambazova, Boryana. . "Welcome to Bulgaria, Where the Ukraine War Is NATO's Fault." *Politico*. 8 June 2022. <https://www.politico.eu/article/bulgaria-ukraine-russia-war-nato-fault/>.

Edmunds, Michael. , and Tony. Freeth . "Using Computation to Decode the First Known Computer." *Computer* 44, no. 7 (2011): 32.

Edunov, Sergey. , Carlos. Diuk , Ismail Onur. Filiz , Smriti. Bhagat , and Moira. Burke . "Three and a Half Degrees of Separation." *Research at Facebook* (blog) 694, February 4, 2016. <https://research.fb.com/three-and-a-half-degrees-of-separation/>. Retrieved 9 September 2022.

Elgammal, Ahmed. . "How Artificial Intelligence Completed Beethoven's Unfinished Tenth Symphony." *Smithsonian Magazine*, 24 September 2021.

Elisha, Ety. , Josh. Guetzkow , Yaffa Shir-Raz, and Natti. Ronel . "Retraction of Scientific Papers: The Case of Vaccine Research." *Critical Public Health* 32, no. 4 (2022): 533–542.

Emanuel, Ryan E. , and David E. Wilkins . "Breaching Barriers: The Fight for Indigenous Participation in Water Governance." *Water* 12, no. 8 (2020): (2113).

Evans, David K. , Maryam. Akmal , and Pamela. Jakiela . "Gender Gaps in Education: The Long View." *IZA, Journal of Development and Migration* 12, no. 1 (2021): 27.

Faciejew, Michael. and Ricardo. Roque . "The Violence of Collections." 22 Sep 2021. *The Order of Multitudes Atlas Encyclopedia*. <https://orderofm.com/conversation/the-violence-of-collections/>

Fallis, Don. , Jonathan. Furner , Kay. Mathiesen , and Allen. Renear . "Philosophy and Information Science: The Basics." In *Proceedings of the American Society for Information Science and Technology* 43, no. 1 (2006): 1–4.

Fass, Craig. , Mike. Ginelli , and Brian. Turtle . *Six Degrees of Kevin Bacon*. New York, NY: Plume Books, 1996.

Faizi, Fatima. , "Afghan Students Run Underground Book Club to Keep Dreams Alive." *Al Jazeera*, 9 May 2022.

Fenlon, Katrina. , Megan. Senseney , Maria. Bonn , and Janet. Swatscheno . "Humanities Scholars and Library-Based Digital Publishing: New Forms of Publication, New Audiences, New Publishing Roles." *Journal of Scholarly Publishing* 50, no. 3 (2019): 159–182.

Fernández, José María Pérez. , and Edward. Wilson-Lee . *Hernando Colon's New World of Books: Toward a Cartography of Knowledge*. New Haven, CT: Yale University Press, 2021.

Fincke, Jeanette C. "The British Museum's Ashurbanipal Library Project." *Iraq* 66 (2004): 55–60.

Fiormonte, Domenico. , Sukanta. Chaudhuri , and Paola. Ricaurte , eds. *Global Debates in the Digital Humanities*. Minneapolis, MN: University of Minnesota Press, 2022.

fitzNigel, Richard. , translator. *Dialogus de Scaccario*. In *The Dialogue of the Exchequer*, edited by Emilie. Amt and S.D. Church . Oxford: Oxford University Press, 2007.

Flanders, Julia. , and Elli. Mylonas . "Digital Humanities." *Encyclopedia of Library and Information Sciences*, edited by Marcia J. Bates and Mary Niles Maack. Boca Raton, FL: CRC Press, 2017, 1286–1297.

Floridi, Luciano. . *The Philosophy of Information*. Oxford: Oxford University Press, 2013.

Fölster, Max Jakob. . "Libraries and Archives in the Former Han Dynasty (206 BCE–9 CE): Arguing for a Distinction." In *Manuscripts and Archives: Comparative Views on Record-Keeping*, edited by Sabine. Kienitz , Michael. Friedrich , Christian. Brockmann and Alessandro. Bausi . Berlin: de Gruyter, 2018, 201–230.

Ford, Anne. . "Bringing Harassment out of the History Books." *American Libraries* 49, no. 6 (2018).

Forrest, Anne. , David. Fazio , Jamie. Knight , Tim. Harshbarger , Jeanine. Lineback , Sandeep. Tirumangalam , Birkir. Gunnarsson , Laura. Goslin (moderator). *Virtual Panel: How Persons with Disabilities Use the Web*. Deque Systems, Inc., 2017. <https://accessibility.deque.com/on-demand-how-persons-with-disabilities-use-the-web>.

Foucault, Michel. . *The Order of Things*. London: Routledge, 2005.

Fowler, Geoffrey. . "How Does Google's Monopoly Hurt You?" *The Washington Post*, 20 October 2020.

Fränti, Pasi. , Sami. Sieranoja , Katja. Wikström , and Tiina. Laatikainen . "Clustering Diagnoses From 58 Million Patient Visits in Finland between 2015 and 2018." *JMIR Medical Informatics* 10, no. 5 (2022): e35422.

Freeth, Tony. , Alexander. Jones , John M. Steele , and Yanis. Bitsakis . "Calendars with Olympiad Display and Eclipse Prediction on the Antikythera Mechanism." *Nature* 454, no. 7204 (2008): 614–617.

Freeth, Tony. , Yanis. Bitsakis , Xenophon. Moussas , John H. Seiradakis , Agamemnon. Tselikas , Helen. Mangou , Mary. Zafeiropoulou et al. "Decoding the Ancient Greek Astronomical Calculator Known as the Antikythera Mechanism." *Nature* 444, no. 7119 (2006): 587.

Freeth, Tony. . "An Ancient Greek Astronomical Calculation Machine Reveals New Secrets." *Scientific American*, 1 January 2022.

Freeth, Tony. . "Building the Cosmos in the Antikythera Mechanism." *Proceedings of Science 170 Antikythera and SKA*, (2013): 1–16. <https://doi.org/10.22323/1.170.0018>.

Furner, Jonathan. . "Definitions of 'Metadata': A Brief Survey of International Standards." *Journal of the Association for Information Science and Technology* 71, no. 6 (2020): E33–E42.

Gadbury, Matt. , and H. Chad Lane . "Mining for STEM Interest Behaviors in Minecraft." In *International Conference on Artificial Intelligence in AI*, edited by Maria Mercedes. Rodrigo , Noburu. Matsuda , Alexandra I. Cristea and

Vania. Dimitrova , 236–239. Cham: Springer, 2022.

Gainer, Kim D. "Fair Use and Digitization: Google Prevails in the Latest Court Ruling." *The CCCC-IP Annual: Top Intellectual Property Developments of 2013*, edited by Ratliff, Clancy (2014): 37.

<https://escholarship.org/uc/item/6ng4h3mj>. Retrieved 22 September 2022.

Gamsu, Sol. . "Why Are Some Children Worth More Than Others? The Private-State School Funding Gap in England." In *Common-Wealth Thinktank*. Durham University, 2021. <https://www.common-wealth.co.uk/reports/why-are-some-children-worth-more-than-others>.

Gardner, Sue. . "Nine Reasons Women Don't Edit Wikipedia (in Their Own Words)." Sue Gardner's Blog. 19 February 2011.

<https://suegardner.org/2011/02/19/nine-reasons-why-women-dont-edit-wikipedia-in-their-own-words/>.

Gary, Jesse. . "Photo of Girls Using Taco Bell Wifi Become Symbol of Digital Divide." KTVU FOX, 31 August 2020. <https://www.ktvu.com/news/photo-of-girls-using-taco-bell-wifi-becomes-symbol-of-digital-divide>.

Gaubert, Julie. . "After More Than Two Centuries, Beethoven's 10th Symphony Has Been Completed by an AI." *EuroNews.next*, 14 October 2021.

Gebru, Timnit. and Margaret. Mitchell . "We Warned Google That People Might Believe AI Was Sentient. Now It's Happening." *The Washington Post*, 17 June 2022.

"Gender Bias on Wikipedia." Wikipedia. Retrieved 29 May 2022.

https://en.wikipedia.org/wiki/Gender_bias_on_Wikipedia.

Georgia, Wells. , Jeff. Horwitz and Deepa. Seetharaman , "Facebook Knows Instagram Is Toxic for Teen Girls, Company Documents Show." *Wall Street Journal*, September 14, 2021.

Gerdon, Frederic. , Helen. Nissenbaum , Ruben L. Bach , Frauke. Kreuter , and Stefan. Zins . "Individual Acceptance of Using Health Data for Private and Public Benefit: Changes during the COVID-19 Pandemic." *Harvard Data Science Review*, Special Issue 1: COVID-19: Unprecedented Challenges and Chances (2021).

Giannoumis, G. Anthony. and Michael Ashley. Stein . "Conceptualizing Universal Design for the Information Society through a Universal Human Rights Lens." *International Human Rights Law Review* 8, no. 1 (2019): 38–66.

Goldstein, Dana. . "Two States. Eight Textbooks. Two American Stories." *The New York Times*, 12 January 2020.

Gong, S. , Loy, C. C. , & Xiang, T. "Security and Surveillance." In *Visual Analysis of Humans*. Cham: Springer, 2011: 455–472.

"Google Search | Overview." Retrieved 15 June 2022.

<https://www.google.com/search/howsearchworks/>.

Graham, Sarah. . "Scientists Win Latest Ruling in Kennewick Man Case." *Scientific American*. 6 February 2004.

Granovetter, Mark S. "The Strength of Weak Ties." *American Journal of Sociology* 78, no. 6 (1973): 1360–1380.

Gray, LaVerne. . "Naomi Willie Pollard Dobson: A Pioneering Black Librarian." *Libraries: Culture, History, and Society* 6, no. 1 (2022): 1–20.

Green, Sara Jean. . "A Wrong Had Finally Been Righted': Tribes Bury Remains of Ancient Ancestor Known as Kennewick Man." *Seattle Times*, 19 February 2017.

Greitens, S. C. "Dealing with Demand for China's Global Surveillance Exports." *Brookings Institution Global China Report* (2020).

Greitens, S. C. "Surveillance, Security, and Liberal Democracy in the Post-COVID World." *International Organization*, 74(S1) (2020): E169–E190.

Grigar, Dene. . "Challenges to Archiving and Documenting Born-Digital Literature: What Scholars, Archivists, and Librarians Need to Know." *Electronic Literature as Digital Humanities: Contexts, Forms, and Practices*, edited by Dene Grigar and James O'Sullivan (2021): 237–245.

Guerrini, Christi J. , Mary A. Majumder , Meaganne J. Lewellyn , and Amy L. McGuire . "Citizen Science, Public Policy." *Science* 361, no. 6398 (2018): 134–136.

Gundaker, Grey. . "Hidden Education among African Americans during Slavery." *Teachers College Record* 109, no. 7 (2007): 1591–1612.

Guo, Jinchi. , and Jie. Huang . "Information Literacy Education during the Pandemic: The Cases of Academic Libraries in Chinese Top Universities." *Journal of Academic Librarianship* 47, no. 4 (2021): 102363.

Gupta, Aditi. , Sarbjit Kaur. Randhawa , and In-In. Po . "Reading Habits, Library Perceptions and Library Usage of the South Asian Population in British Columbia, Canada." *Public Library Quarterly* 41, no. 5 (2021): 456–484.

Gupta, Meghanlata. . "Regaining Control: Indigenous-Owned and Operated Archives." Retrieved 11 June 2022.
<https://indiancountrytoday.com/news/regaining-control-indigenous-owned-and-operated-archives>.

Hammer, Joshua. . "The Brave Sage of Timbuktu: Abdel Kader Haidara." *The Innovators Project*. National Geographic. 2014.
<https://www.nationalgeographic.com/culture/article/140421-haidara-timbuktu-manuscripts-mali-library-conservation>.

Hansson, Joacim. . *Educating Librarians in the Contemporary University: An Essay on iSchools and Emancipatory Resilience in Library and Information Science*. Sacramento, CA: Library Juice Press, 2019.

Hapke, Thomas. . "Wilhelm Ostwald, 'The 'Brücke' (Bridge), and Connections to Other Bibliographic Activities at the Beginning of the Twentieth Century." *Information Today*, 29, no. 4, 1999: 142.

Harding, Luke. and Harriet. Sherwood . "Ukrainians in Race to Save Cultural Heritage." *The Guardian*, 9 March 2022.

Hargittai, Eszter. ; Walejko, Gina. ; Gina. Walejko . "The Participation Divide: Content Creation and Sharing in the Digital Age." *Information, Communication & Society* 11, no. 2 (2008): 20.

Hasinoff, Amy Adele. , and Tamara. Shepherd . "Sexting in Context: Privacy Norms and Expectations." *International Journal of Communication* 8 (2014): 24.

Haunui-Thompson, Shannon. . "Whanganui to Gain Legal Personhood." *New Zealand Radio*. 16 March 2017.

Heer, J. , M. Bostock and V. Ogievetsky . "A Tour through the Visualization Zoo." *ACM Queue* (2010) 53(6), 59–67.

Heitlinger, Sara. , Lara. Houston , Alex. Taylor , and Ruth. Catlow .
 “Algorithmic Food Justice: Co-Designing More-Than-Human Blockchain
 Futures for the Food Commons.” In *Proceedings of the 2021 CHI
 Conference on Human Factors in Computing Systems* (2021): 1–17.
<https://doi.org/10.1145/3411764.3445655>.

Helen, Nissenbaum. . “Shaping the Web: Why the Politics of Search Engines
 Matters.” *The Information Society* 16 (3) (2000): 169–185.

Helton, Laura E. “On Decimals, Catalogs, and Racial Imaginaries of
 Reading.” *PMLA* 134, no. 1 (2019): 105.

Henry, Ella. , and Hone. Pene . “Kaupapa Maori: Locating Indigenous
 Ontology, Epistemology and Methodology in the Academy.” *Organization* 8,
 no. 2 (2001): 234–242.

Herbrechter, Stefan. , and Ivan. Callus . *Cy-Borges: Memories of the
 Posthuman in the Work of Jorge Luis Borges*. Lewisburg, PA: Bucknell
 University Press, 2009.

Hermon, P. and H.C. Relyea , “Information Policy.” In *Encyclopedia of library
 and information science*. New York: Marcel Dekker, Inc., 2003.

Herring, Susan C. , J. Reagle , Justine. Cassell , Terri. Oda , Anna. North , J.
 West , and M. Ranga . “Where Are the Women in Wikipedia.” *The New York
 Times* (February 2, 2011).
[https://www.nytimes.com/roomfordebate/2011/02/02/where-are-the-women-
 in-wikipedia](https://www.nytimes.com/roomfordebate/2011/02/02/where-are-the-women-in-wikipedia). Retrieved 22 September, 2022.

Higgins, Sarah. . “Digital Curation: The Development of a Discipline within
 Information Science.” *Journal of Documentation* 74, no. 6, (2018).
<https://doi.org/10.1108/JD-02-2018-0024>.

Hill, Sharon. . “U of W Prof Gets ‘Indiana Jones Feeling’ with Discovery of
 Historic Manuscript.” *Windsor Star*, 12 April 2019.

Hinchcliffe, L. J. “From Paywall to Datawall.” *The Scholarly Kitchen* (2018).
<https://scholarlykitchen.sspnet.org/2018/10/11/from-paywall-to-datawall/>.

Hjørland, Birger. , and Hanne. Albrechtsen . “Toward a New Horizon in
 Information Science: Domain-Analysis.” *Journal of the American Society for
 Information Science* 46, no. 6 (1995): 400–425.

Hjørland, Birger. . “Knowledge Organization (KO).” *KO Knowledge
 Organization* 43, no. 6 (2016): 475–484.

Hoiem, Elizabeth Massa. . “The Progress of Sugar: Consumption as
 Complicity in Children's Books about Slavery and Manufacturing,
 1790–2015.” *Children's Literature in Education*, 52, no. 2 (2021): 162–182.

Holroyd, Carin. . “Digital Content Promotion in Japan and South Korea:
 Government Strategies for an Emerging Economic Sector.” *Asia & the
 Pacific Policy Studies* 6, no. 3 (2019): 290–307.

Hopping, David. . “Modeling Hope: Boundary Objects and Design Patterns in
 a Heartland Heterotopia.” In *Deciding Where to Live: Information Studies on
 Where to Live in America*, edited by Ocepek, Melissa G. , and William.
 Aspray . Lanham, MD: Rowman & Littlefield, 2020: 211–235.

Houston, Ronald D. and Glynn. Harmon . “Vannevar Bush and Memex.”
Annual Review of Information Science and Technology 41, no. 1 (2007):
 55–92.

“How Google Search Organizes Information.” Retrieved 28 May 2022.
<https://www.google.com/search/howsearchworks/how-search->

works/organizing-information/.

Hughes, Sallie. , and Yulia. Vorobyeva . “Explaining the Killing of Journalists in the Contemporary Era: The Importance of Hybrid Regimes and Subnational Variations.” *Journalism* 22, no. 8 (2021): 1873–1891.

Huttunen, Aira. , Lottamari. Kähkönen , Heidi. Enwald , and Terttu. Kortelainen . “Embodied Cognition and Information Experiences of Transgender People.” *Information Research* 24 no. 4 (2019).

<http://InformationR.net/ir/24-4/colis/colis1940.html>. Retrieved 22 September 2022.

Huybrechts, Liesbeth. , Henric. Benesch , and Jon. Geib . “Co-Design and the Public Realm.” *CoDesign* 13, no. 3 (2017): 145–147.

Ibekwe-Sanjuan, Fidelia. . “Whither Information Science in France?” In *International Perspectives on the History of Information Science and Technology*. Baltimore, MD: Association of the American Society for Information Science and Technology, 2012: 83–95.

“IFLA Statement on Privacy in the Library Environment.” International Federation of Library Associations and Institutions. Retrieved 4 June 2022. <https://www.ifla.org/publications/ifla-statement-on-privacy-in-the-library-environment/>.

“Indigenous Peoples and Local Communities Portal.” WIPO. <https://www.wipo.int/tk/en/indigenous/>. Retrieved 2 June 2022.

“International Council of Museums | About.” ICOM. <https://icom.museum/en/about-us/>. Retrieved 10 June 2022.

“International Press Institute.” Retrieved 3 June 2022. <https://ipi.media/>.

“Internet Hall of Fame Pioneer: J.C.R. Licklider.” Internet Hall of Fame. <https://www.internethalloffame.org/inductees/jcr-licklider>. Retrieved 14 June 2022.

Isopahkala-Bouret, Ulpukka. , Mikael. Börjesson , Dennis. Beach , Nina. Haltia , Jón Torfi. Jónasson , Annukka. Jauhiainen , Arto. Jauhiainen , Sonja. Kosunen , Hanna. Nori , and Agnete. Vabø . “Access and Stratification in Nordic Higher Education. A Review of Cross-Cutting Research Themes and Issues.” *Education Inquiry* 9, no. 1 (2018): 142–154.

Jacobs, Margaret D. “Indian Boarding Schools in Comparative Perspective: The Removal of Indigenous Children in the United States and Australia, 1880–1940.” Digital Commons at the University of Nebraska, Lincoln (2006). <https://digitalcommons.unl.edu/historyfacpub/20/>.

James Murdoch. , Mark. Bauerlein , Marie. Halverson ; Natalie. Morrissey ; and Esther. Galadima . “How Do We Read? Let’s Count the Ways: Comparing Digital, Audio, and Print-Only Readers.” National Endowment for the Arts Report (2020): 16. <https://www.arts.gov/sites/default/files/How%20Do%20We%20Read%20report%202020.pdf>. Retrieved 23 September 2022.

Justin Wm, Moyer. . “Fighting Racial Bias with an Unlikely Weapon: Footnotes.” *The Washington Post*, 18 January 2022.

Kamran, Qeis. , and Berit. Schumann . “Towards an Effective Rapprochement between Artificial Intelligence and Medical and Pharmaceutical Research.” *Modern Approaches in Drug Designing* 3, no. 4 (2022): 1–4.

Karen, Hao. , "The Facebook Whistleblower Says Its Algorithms Are Dangerous. Here's Why." MIT Technology Review. Retrieved 7 June 2022. https://canvas.illinois.edu/courses/16422/discussion_topics/125024.

Karinthy, Frigyes. . "Chain-links." In Everything Is Different. Translated by Adam. Makkai . Edited by Enikő. Jankó . Public Domain, http://vadeker.net/articles/Karinthy-Chain-Links_1929.pdf. Retrieved 22 September 2022. Figyes Karinthy, "Láncszemek," in Minden másképpen van" (Budapest: Atheneum Irodalmi és Nyomdai R.-T, 1929): 85–91.

Kelly, Diann Cameron. , and Rani. Varghese . "Four Contexts of Institutional Oppression: Examining the Experiences of Blacks in Education, Criminal Justice and Child Welfare." Journal of Human Behavior in the Social Environment 28, no. 7 (2018): 874–888.

Kennan, Mary Anne. , and Jessie. Lymn . "Where Is the I(nformation) in GLAM? Education, Knowledge and Skill Requirements of Professionals Working in GLAM Sector Institutions." Journal of the Australian Library and Information Association 68, no. 3 (2019): 236–253.

Khademian, Mahdi. , and Morteza. Kokabi . "Library Thing Social Tags versus Library of Congress Subject Headings: A Literature Review." Library and Information Science Research 8, no. 1 (2018): 313–335.

Khalil, Ashraf. , Soha Glal. Ahmed , Asad Masood. Khatkhat , and Nabeel. Al-Qirim . "Investigating Bias in Facial Analysis Systems: A Systematic Review." IEEE Access 8 (2020): 130751–130761.

Kimmerer, Robin. . Braiding Sweetgrass: Indigenous Wisdom, Scientific Knowledge and the Teachings of Plants. Minneapolis, MN: Milkweed editions, 2013: 208.

Kirchhoff, Thomas. , Werner. Schweibenz , and Jörn. Sieglerschmidt . "Archives, Libraries, Museums and the Spell of Ubiquitous Knowledge." Archival Science 8, no. 4 (2008): 256.

Kline, Ronald R. "What Is Information Theory a Theory of? Boundary Work among Information Theorists and Information Scientists in the United States and Britain during the Cold War." In The History and Heritage of Scientific and Technological Information Systems, edited by W.B. Rayward and M.E. Bowden . Medford, NJ: American Society of Information Science and Technology and the Chemical Heritage Foundation, 2004: 15–28.

Knowlton, Steven A. "Three Decades since Prejudices and Antipathies: A Study of Changes in the Library of Congress Subject Headings." Cataloging & Classification Quarterly 40, no. 2 (2005): 123–145.

Knox, Emily J.M. . "Opposing Censorship in Difficult Times." The Library Quarterly 87, no. 3 (2017).

Knox, Emily J.M. . Book Banning in 21st-Century America. Lanham, MD: Rowman & Littlefield, 2015.

Kochen, Manfred. and Ithiel. da Sola Pool . "Contacts and Influences." Social Networks 1 (1978): 5–51.

Kokshagina, Olga. . "Open Covid-19: Organizing an Extreme Crowdsourcing Campaign to Tackle Grand Challenges." R&D Management 52, no. 2 (2022): 206–219.

Kostka, Genia. , Léa. Steinacker , and Miriam. Meckel . "Between Security and Convenience: Facial Recognition Technology in the Eyes of Citizens in China, Germany, the United Kingdom, and the United States." Public

Understanding of Science 30, no. 6 (2021): 671–690.

Kramer, Jessica M. , John C. Kramer , Edurne. García-Iriarte , and Joy. Hammel . “Following through to the End: The Use of Inclusive Strategies to Analyse and Interpret Data in Participatory Action Research with Individuals with Intellectual Disabilities.” *Journal of Applied Research in Intellectual Disabilities* 24, no. 3 (2011): 263–273.

Kuhlthau, Carol C. “Inside the Search Process: Information Seeking from the User's Perspective.” *Journal of the American Society for Information Science* 42, no. 5 (1991): 361–371.

La Barre, Kathryn. . “Facet Analysis.” *Annual Review of Information Science and Technology* 44, no. 1 (2010): 243.

Lahti, Leo. , Jani. Marjanen , Hege. Roivainen , and Mikko. Tolonen . “Bibliographic Data Science and the History of the Book (c. 1500–1800).” *Cataloging & Classification Quarterly* 57, no. 1 (2019): 5–23.

Lancaster, F. W. *Indexing and Abstracting in Theory and Practice*. 2nd ed. Champaign, IL: University of Illinois Graduate School of Library and Information Science, 1998.

Lauren, Poole. . “Bromley's Model of the Antikythera Mechanism.” *Powerhouse*, 1 November 2017, <https://www.maas.museum/inside-the-collection/2017/11/01/bromleys-model-of-the-antikythera-mechanism/>.

Lawrimore, Erin. . “Margaret Cross Norton: Defining and Redefining Archives and the Archival Profession.” *Libraries & the Cultural Record* 44, no. 2 (2009): 186.

Le Deuff, Olivier. , and Arthur. Perret . “Hyperdocumentation: Origin and Evolution of a Concept.” *Journal of Documentation* 75, no. 6 (2019): 6.

Leazer, Gregory H. , and Robert. Montoya . “The Politics of Knowledge Organization: Introduction to the Special Issue.” *Knowledge Organization* 47, no. 5 (2020): 367–371.

Lee, Ronan. . “Extreme Speech in Myanmar: The Role of State Media in the Rohingya Forced Migration Crisis.” *International Journal of Communication* 13 (2019): 3202–3224.

Leibold, James. . “Surveillance in China's Xinjiang Region: Ethnic Sorting, Coercion, and Inducement.” *Journal of Contemporary China* 29, no. 121 (2020): 46–60.

“Leighton Linslade Town Council EWHC 760 (Admin).” 15 February 2019, <http://www.bailii.org/ew/cases/EWHC/Admin/2019/760.html>.

Lember, Heather. , Suzanne. Lipkin , and Richard Jung. Lee . “Radical Cataloging: From Words to Action.” *Urban Library Journal* 19, no. 1 (2013): 7.

Lemov, Michael R. , and Nate. Jones . “John Moss and the Roots of the Freedom of Information Act: Worldwide Implications.” *Sw J Int'l L.* 24 (2018): 1.

Levie, Francoise. (director). *The Man Who Wanted to Classify the World*. New York: Filmmakers Library, 2004.

Library of Babel (website). Retrieved 17 May 2022. <https://libraryofbabel.info/About.html>.

Licklider, Joseph CR. , and Robert W. Taylor . “The Computer as a Communication Device.” *Science and Technology* 76, no. 2 (1968): 1–3.

Lin, Paul J. A Translation of Lao-tzu's Tao Te Ching and Wang Pi's Commentary. Ann Arbor, MI: University of Michigan Press, 2020.

Llewellyn, Clare. , Laura. Cram , Robin L. Hill , and Adrian. Favero . "For Whom the Bell Tolls: Shifting Troll Behaviour in the Twitter Brexit Debate." *JCMS: Journal of Common Market Studies* 57, no. 5 (2019): 1148–1164.

Logan, Wayne A. , and Andrew Guthrie. Ferguson . "Policing Criminal Justice Data." *Minnesota Law Review* 101 (2016): 541.

Longfellow, Henry Wadsworth. . "Santa Filomena." *The Atlantic Monthly*, November 1857.
<https://www.theatlantic.com/magazine/archive/1857/11/santa-filomena/531180/>.

"Local, State, and National Laws and Constitutional Frameworks, Uganda." Right to Nature Law Library, Center for Democratic and Environmental Rights. <https://www.centerforenvironmentalrights.org/rights-of-nature-law-library>. Retrieved 19 May 2022.

"Looking at One's Self through the Eyes of Others: W. E. B. Du Bois's Photographs for the 1900 Paris Exposition." *African American Review* 34, no. 4 (Winter 2000): 581–599.

Lu, Wei. , and Max. Aiken . "Origins and Evolution of Chinese Writing Systems and Preliminary Counting Relationships." *Accounting History* 9, no. 3 (2004): 25–51.

Lueg, C.P. "The Missing Link: Information Behavior Research and Its Estranged Relationship with Embodiment." *Journal of the Association for Information Science and Technology*, 66(12) (2015): 2704–2707.

Luo, Lili. . "Being Interdisciplinary: A Look into the Background and Experiences of iSchool Faculty Members." *LIBRES: Library and Information Science Research Electronic Journal* 23, no. 2 (2013): 1–20.

Łysak, Tomasz. . "Vlogging Auschwitz: New Players in Holocaust Commemoration." *Holocaust Studies* (2021): 1–26.

Maack, Mary Niles. . "The Lady and the Antelope: Suzanne Briet's Contribution to the French Documentation Movement." *Library Trends* 52 (4) (2004): 737.

Mace, Ronald L. "Universal Design in Housing." *Assistive Technology* 10, no. 1 (1998): 21–28.

Madden, Andrew D. , Jared. Bryson , and Joe. Palimi . "Information Behavior in Pre-Literate Societies." In *New Directions in Human Information Behavior*, edited by Amanda Spink and Charles Cole. Springer, 2006: 8–9.

Madubuike-Ekwe, N. J. , and Joseph N. Mbadugha . "Obstacles to the Implementation of the Freedom of Information Act, 2011 in Nigeria." *Nnamdi Azikiwe University Journal of International Law and Jurisprudence* 9, no. 2 (2018): 96–109.

Maemura, Emily. , Nicholas. Worby , Ian. Milligan , and Christoph. Becker . "If These Crawls Could Talk: Studying and Documenting Web Archives Provenance." *Journal of the Association for Information Science and Technology* 69, no. 10 (2018): 1223–1233.

Maggie, Puniewska. . "Science Has a Sharing Problem." *The Atlantic*, 15 December 2014.
<https://www.theatlantic.com/health/archive/2014/12/scientists-have-a-sharing-problem/383061/>.

Magee, Rachel M. , "Teen Social Media Practices and Perceptions of Peers." *The Journal of Research on Libraries and Young Adults*, 10, no. 3 (2019): 1–121.

Magi, Trina J. "Confidentiality: Best Practices Protecting Library Patron Confidentiality: Checklist of Best Practices." <https://www.ila.org/advocacy/making-your-case/privacy/confidentiality-best-practices>. Retrieved 22 June 2022.

Magi, Trina J. and Martin. Garnar , *Intellectual Freedom Manual*. 9th ed. Chicago, IL: American Library Association and Office for Intellectual Freedom, 2015.

Mak, Bonnie. , and Julia. Pollack . "The Performance and Practice of Research in a Cabinet of Curiosity: The Library's Dead Time." *Art Documentation: Journal of the Art Libraries Society of North America* 32, no. 2 (2013): 202–221.

Mak, Bonnie. . "Wood Libraries: Knowing with Wood." *Caxtonian*, May–June, 2021.

Mak, Bonnie. . *How the Page Matters*. Toronto: University of Toronto Press, 2011.

Māni, Dunlop. . "University Academics' Claim Mātauranga Māori 'Not Science' Sparks Controversy." *Radio New Zealand*, 28 July 2021.

Mansky, Jackie. . "W.E.B. Du Bois' Visionary Infographics Come Together for the First Time in Full Color." *Smithsonian Magazine*, 15 November 2018. <https://www.smithsonianmag.com/history/first-time-together-and-color-book-displays-web-du-bois-visionary-infographics-180970826/>.

Marmier, Auriane. . "The Publication of Open Government Data." *Dance Your PhD* 2022. <https://www.youtube.com/watch?v=gOpK0TZ61bY>. Retrieved 4 June 2022.

Martin, Nicole. . "Indigenous Rights: An Analysis of Intellectual Property Protections." *Am. U. Intell. Prop. Brief* 13 (2021): 33.

Matejka, Justin. and George. Fitzmaurice . "Same Stats, Different Graphs: Generating Datasets with Varied Appearance and Identical Statistics through Simulated Annealing." In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (2017): 1290–1294. <https://doi.org/10.1145/3025453.3025912>.

Max, Roser. and Esteban. Ortiz-Ospina . "Literacy: Data Quality Challenges and Limitations." In *Our World in Data: Literacy*. Retrieved 31 May 2022. <https://ourworldindata.org/literacy>.

McDonough, Jerome P. , Robert. Olendorf , Matthew. Kirschenbaum , Kari. Kraus , Doug. Reside , Rachel. Donahue , Andrew. Phelps , Christopher. Egert , Henry. Lowood , and Susan. Rojo . *Preserving Virtual Worlds Final Report*. 2010. <http://hdl.handle.net/2142/17097>.

McDonough, Jerome Patrick. . *Under Construction: The Application of a Feminist Sociology to Information Systems Design*. Berkeley, CA: University of California, 2000.

McDowell, Kate. , Nicole. Cooke , Janice. Del Negro , Beth. Patin , and Curtis. Tenney . "Storytelling and/as Resilience." *Political Economy of the Information Society* (2021).

McDowell, Kate. . "Paradoxes of Storytelling in Librarianship." *Journal of New Librarianship* 3 (2018): iv.

McDowell, Kate. . "Storytelling Wisdom: Story, Information, and DIKW." *Journal of the Association for Information Science and Technology* 72, no. 10 (2021): 1223–1233.

McKenzie, Pete. . "Can Ancient Maori Knowledge Aid Science? Ask These Freshwater Crayfish." *New York Times*, 1 June 2022.

McKitterick, David. . "That Woman! Studies in Irish Bibliography: A Festschrift for Mary Paul Pollard." *The Library: The Transactions of the Bibliographical Society* 7, no. 2 (2006): 210–211.

Menking, Amanda. , Ingrid. Erickson , and Wanda. Pratt . "People Who Can Take It: How Women Wikipedians Negotiate and Navigate Safety." *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, Paper No.: 472 (Glasgow, Scotland: SIGCHI 2019): 1–14. <https://doi.org/10.1145/3290605.3300702>.

Metcalf, John. . *Subject Classifying and Indexing of Libraries and Literature*. Sydney, NSW: Angus and Robertson, 1959.

Meyer, Eric. and Sara. Wachter-Boettcher . *Design for Real Life*. New York: A Book Apart, 2016.

Milgram, Stanley. . "The Small World Problem." *Psychology Today* 2, no. 1 (1967): 60–67.

Miller, Jessie. . "Tracking the Tracing: A Global Investigation of Privacy Issues in the Age of COVID-19." *Dartmouth Undergraduate Journal of Politics, Economics and World Affairs* 1, no. 3 (2021): 2.

Minocher, Xerxes. , and Caelyn. Randall . "Predictable Policing: New Technology, Old Bias, and Future Resistance in Big Data Surveillance." *Convergence* 26, no. 5–6 (2020): 1108–1124.

Mirza, Rafia. , and Maura. Seale . "Who Killed the World? White Masculinity and the Technocratic Library of the Future." *Topographies of Whiteness: Mapping Whiteness in Library and Information Science* 1 (2017): 172.

Mizrachi, Diane. . "Undergraduates' Academic Reading Format Preferences and Behaviors." *Journal of Academic Librarianship* 41, no. 3 (2015): 301–311.

Montenegro, María. . "Subverting the Universality of Metadata Standards: The TK Labels as a Tool to Promote Indigenous Data Sovereignty." *Journal of Documentation* (2019): 737.

Montesi, Michela. . "Human Information Behavior during the Covid-19 Health Crisis. A Literature Review." *Library & Information Science Research* 43, no. 4 (2021): 101122.

"Monumental Broadband Legislation by Senator Lena Gonzalez and Assemblymember Cecilia Aguiar-Curry Signed into Law." 8 October 2021. <https://sd33.senate.ca.gov/news/2021-10-08-monumental-broadband-legislation-senator-lena-gonzalez-and-assemblymember-cecilia>.

Mote, Lionel JB. . "Reasons for the Variations in the Information Needs of Scientists." *Journal of Documentation* (1962). Cited in Wilson, Human, 51.

Mueller, Jason. , Genevieve. Perez , "'Deepfake' Technology: Very Real Marketing Value ... and Risks." *National Law Review* (2020).

"Museum of Broken Relationships." Retrieved 24 May 2022. <https://brokenships.com/visit>.

Nahotko, Marek. . "Knowledge Organization Affordances in a Faceted Online Public Access Catalog (OPAC)." *Cataloging & Classification Quarterly* 60, no. 1 (2022): 86–111.

Naudé, Gabriel. . *Advis pour dresser une bibliotheque*. 1627. Translated by John Evelyn as *Instructions Concerning Erecting of a Library*. London, 1661. Houghton, Mifflin at the Riverside Press, 1903.

"Native American Graves Protection and Repatriation Act." National Park Service. <https://www.nps.gov/subjects/nagpra/index.htm>. Retrieved 9 June 2022.

Nesi, Jacqueline. , Taylor A. Burke , Alexandra H. Bettis , Anastacia Y. Kudinova , Elizabeth C. Thompson , Heather A. MacPherson , Kara A. Fox et al. "Social Media Use and Self-Injurious Thoughts and Behaviors: A Systematic Review and Meta-Analysis." *Clinical psychology review* 87 (2021): 102038.

"Necessary and Proportionate. International Principles on the Application of Human Rights to Communications Surveillance." EFF (2014). Retrieved 22 June 2022. <https://necessaryandproportionate.org/principles/>.

Nightingale, Florence. . "Letter from Nightingale to Queen Victoria, 1863." Florence Nightingale Letters Collection. University of Illinois Chicago Library Special Collections. Retrieved 24 May 2022. https://collections.carli.illinois.edu/digital/collection/uic_fnlc/id/23.

Nightingale, Florence. . *A Contribution to the Sanitary History of the British Army during the Late War with Russia*. John W. Parker, 1859.

Nijdam, Elizabeth Biz. . "Sami-Digital Storytelling: Survivance and Revitalization in Indigenous Digital Games." *New Media & Society* (2021). <https://doi.org/10.1177/14614448211038902>

Nissenbaum, Helen. . "A Contextual Approach to Privacy Online." *Daedalus* 140, no. 4 (2011): 32–48.

Noble, Safiya Umoja. . "Algorithms of Oppression." In *Algorithms of Oppression: How Search Engines Reinforce Racism*. New York: New York University Press, 2018.

Norman, Don. . *The Design of Everyday Things: Revised and Expanded Edition*. New York, NY: Basic Books, 2013: 8–9. See also Lara Fedoroff, host, "Reflections from Design of Everyday Things, with Don Norman." UX Radio (podcast), 10 December 2015. <http://ux-radio.com/2015/12/reflections-design-everyday-things/>.

Nunes, Zita Cristina. . "Remembering the Howard University Librarian Who Decolonized the Way Books Were Catalogued." *The Smithsonian Magazine*, 26 November 2018.

Nunn, Patrick D. , and Nicholas J. Reid . "Aboriginal Memories of Inundation of the Australian Coast Dating from More Than 7000 Years Ago." *Australian Geographer* 47, no. 1 (2016): 11–47.

O'Connor, Karl. , Saltanat. Janenova , and Colin. Knox . "Open Government in Authoritarian Regimes." *International Review of Public Policy* 1, no. 1: 1 (2019): 65–82.

Obermeyer, Ziad. , Brian. Powers , Christine. Vogeli , and Sendhil. Mullainathan . "Dissecting Racial Bias in an Algorithm Used to Manage the Health of Populations." *Science* 366, no. 6464 (2019): 447–453.

O'Brien, Tia. . "From the Archives: Douglas Engelbart's Lasting Legacy, 1999." Silicon Valley.com/The Mercury News, 3 July 2013.

Ocepek, Melissa G. "Sensible Shopping: A Sensory Exploration of the Information Environment of the Grocery Store." *Library Trends* 66, no. 3 (2018): 371–394.

O'Donnell, Daniel Paul. , Barbara. Bordalejo , Padmini Murray. Ray , Gimena. del Rio Riande , and Elena. González-Blanco García . "Boundary Land: Diversity as a Defining Feature of the Digital Humanities." (2016). <http://dh2016.adho.org/abstracts/406>.

Öhman, Carl. , and Luciano. Floridi . "The Political Economy of Death in the Age of Information: A Critical Approach to the Digital Afterlife Industry." *Minds and Machines* 27, no. 4 (2017): 639–662.

Oliphant, Tami. . "A Case for Critical Data Studies in Library and Information Studies." *Journal of Critical Library and Information Studies* 1, no. 1 (2017). <https://doi.org/10.24242/jclis.v1i1.22>.

Olson, Hope A. "Difference, Culture and Change: The Untapped Potential of LCSH." *Cataloging & Classification Quarterly* 29, no. 1–2 (2000): 54–55.

Ong, Walter J. *Orality and Literacy*. London: Routledge, 2013: 10.

"Open Data." National Museum Palace. <https://theme.npm.edu.tw/opendata/>. Retrieved 20 June 2022.

"Open-GLAM: What." <https://openglam.org/what/>. Retrieved 24 May 2022.

"Our Charter." Te Mana Raraunga Maori Data Sovereignty Network. <https://www.temanararaunga.maori.nz/tutohinga>. Retrieved 22 June 2022.

"The Oracle of Bacon." Retrieved 26 May 2022. <http://oracleofbacon.org/>.

"Our Partnership with Menominee Tribal Members on the Rights of the Menominee River." Center for Democratic and Environmental Rights. <https://www.centerforenvironmentalrights.org/news/our-partnership-with-menominee-tribal-members-on-the-rights-of-the-menominee-river>. Retrieved 20 May 2022.

Ortiz, Stephanie M. "'You Can Say I Got Desensitized to It': How Men of Color Cope with Everyday Racism in Online Gaming." *Sociological Perspectives* 62, no. 4 (2019): 572–588.

Osbourne, Sydney Godolphin . *Scutari and Its Hospitals*. Dickinson, 1855.

Ostwald, Wilhelm. . "Scientific Management for Scientists." *Scientific American*, 108 (1913): 5–6.

Otlet, Paul. . *Monde: Essai d'universalisme: Connaissances du monde. Sentiments du monde. Action organiste et plan du monde* (Editiones Mundaneum). D. Van Keerberghen et fils, 1935.

Papadopoulou, Lambrini. , and Theodora A. Maniou . "'Lockdown' on Digital Journalism? Mapping Threats to Press Freedom during the COVID-19 Pandemic Crisis." *Digital Journalism* no. 9 (2021): 1344–1366.

"Parents of Parkland Shooting Victim Joaquin Oliver Bring Son Back Digitally in New Voting Initiative." CBSNews. 2 October 2020.

Patin, B. , Sebastian, M. , Yeon, J. , Bertolini, D. and Grimm, A. "Interrupting Epistemicide: A Practical Framework for Naming, Identifying, and Ending Epistemic Injustice in the Information Professions." *Journal of the Association for Information Science and Technology* 72, no. 10 (2021): 1306–1318.

Pattison, Mary-Ann. . "The Australian Joint Copying Project." *Government Publications Review* 13, no. 3 (1986): 349–353.

Peter Berger . *Invitation to Sociology*. New York, NY: Anchor Books, 1963.

Pettee, Julia. . "A Classification for a Theological Library." *Library Journal* 36 (2011): 623; cited by Walker, Christopher, and Ann Copeland. "The Eye Prophetic: Julia Pettee." *Libraries & The Cultural Record* 44, no. 2 (2009): 162.

Pettee, Julia. . "The Development of Authorship Entry and the Formulation of Authorship Rules as Found in the Anglo-American Code." *The Library Quarterly* 6, no. 3 (1936): 270–290.

Pichumani, Anusha. . "Universal-Design Principles and Heuristic Guidelines." *UX Matters* (2000).
<https://www.uxmatters.com/mt/archives/2020/11/universal-design-principles-and-heuristic-guidelines.php>.

Pickering, Andrew. . *The Mangle of Practice: Time, Agency, and Science*. Chicago, IL: University of Chicago Press, 2010.

Pintar, Judith. . "The Valley between Us: The Meta-Hodology of Racial Segregation in Milwaukee, Wisconsin." In *Deciding Where to Live: Information Studies on Where to Live in America*, edited by Ocepek, Melissa G. , and William. Aspray . Lanham, MD: Rowman & Littlefield Publishers, 2020: 177–210.

"Policy." World Intellectual Property Organization.
<https://www.wipo.int/policy/en/>. Retrieved 2 June 2022.

"Pollard Collection of Children's Books Now Fully Available for Online Searching!" *The Library of Trinity College Dublin News & Alerts*, 19 February 2018. <https://www.tcd.ie/library/news/pollard-collection-of-childrens-books-now-fully-available-for-online-searching/>.

Povroznik, Nadezhda. . "Digital History of Virtual Museums: The Transition from Analog to Internet Environment." in *Proceedings of the Digital Humanities in the Nordic Countries 5th Conference* (2020): 125–136.
<http://ceur-ws.org/Vol-2612/paper9.pdf>. Retrieved 22 September 2022.

Price, W. Nicholson. , and I. Glenn. Cohen . "Privacy in the Age of Medical Big Data." *Nature Medicine* 25, no. 1 (2019): 37–43.

Pritchard, Steve J. , and Alison L. Weightman . "Medline in the UK: Pioneering the Past, Present and Future." *Health Information & Libraries Journal* 22 Supplement 1 (2005): 38–44.

"Racial Bias on Wikipedia." Retrieved 19 June 2022.
https://en.wikipedia.org/wiki/Racial_bias_on_Wikipedia.

Raign, Kathryn R. "Finding Our Missing Pieces—Women Technical Writers in Ancient Mesopotamia." *Journal of Technical Writing and Communication* 49, no. 3 (2019): 338–364.

Ramsey, Andrew. . "X-Ray Tomography of the Antikythera Mechanism." *Proceedings of Science* 170, Antikythera and SKA (2013): 1-12.
<https://doi.org/10.22323/1.170.0022>.

Rasmussen, Morten. , Martin. Sikora , Anders. Albrechtsen , Thorfinn Sand. Korneliussen , J. Víctor. Moreno-Mayar , G. David. Poznik , Christoph PE. Zollikofer et al. "The Ancestry and Affiliations of Kennewick Man." *Nature* 523, no. 7561 (2015): 455–458.

Rayward, W. Boyd. . "Knowledge Organisation and a New World Polity: The Rise and Fall and Rise of the Ideas of Paul Otlet." *Transnational Associations* 55, no. 1–2 (2003): 4–15.

"Relentless Doppelgänger." *Dadabots*, 4 September 2019. <https://www.youtube.com/watch?v=MwtVkpKx3RA>.

Rico, Christophe. . "The Destruction of the Library of Alexandria: A Reassessment." In *The Library of Alexandria: A Cultural Crossroads of the Ancient World*, edited by Christopher. Rico and Anca. Dan . Jerusalem: Polis Institute Press, 2017: 330.

Rieder, Bernhard. . *Engines of Order: A Mechanology of Algorithmic Techniques*. Amsterdam: Amsterdam University Press, 2020.

Riedlmayer, Andras. . "Erasing the Past: The Destruction of Libraries and Archives in Bosnia-Herzegovina." *Review of Middle East Studies* 29, no. 1 (1995): 7–11.

"The Right to Privacy in the Digital Age." *United Nations Digital Library*, 2019. <https://digitallibrary.un.org/record/3837297?In=en>. Retrieved 22 June 2022.

Riley, Jenn. "Seeing Standards: A Visualization of the Metadata Universe." Retrieved 20 May 2022. <http://jennriley.com/metadatamap/>.

Riley, Jenn. . *Understanding Metadata*. Washington, DC: National Information Standards Organization, 2017: 23.

Riordan, Michael. . "Materials for History? Publishing Records as a Historical Practice in Eighteenth- and Nineteenth-Century England." *History of Humanities* 2, no. 1 (2017): 63.

Roberto, K. R. "Inflexible Bodies: Metadata for Transgender Identities." *Journal of Information Ethics* 20, no. 2 (2011): 56.

Robinson, Lyn. , and David. Bawden . "Mind the Gap: Transitions between Concepts of Information in Varied Domains." In *Theories of Information, Communication and Knowledge*. Dordrecht: Springer, 2014: 121–141.

Robinson, Lyn. , Ernesto. Priego , and David. Bawden . "Library and Information Science and Digital Humanities: Two Disciplines, Joint Future?" In Christian. Schlögl , Christian. Wolff , Franjo. Pehar , editors, *Re-Inventing Information Science in the Networked Society*. Glückstadt: Hülsbusch, 2015.

Rosenfeld, Louis. , Peter. Morville , and Jorge. Arango . *Information Architecture: For the Web and Beyond*. Sebastopol, CA: O'Reilly, 2015.

Ross, Burley. . "The Yalivshchyna Burial Site: Mass Graves after Russian Invasion." *Centre for Information Resilience*. 10 April 2022. <https://www.infores.org/post/the-yalivshchyna-burial-site-mass-graves-after-russian-invasion>.

Roth, Robert E. "Cartographic Design as Visual Storytelling: Synthesis and Review of Map-Based Narratives, Genres, and Tropes." *The Cartographic Journal* 58, no. 1 (2021): 83–114.

Ryan, Mac. and Cecilia. Kang . "Whistle-Blower Says Facebook 'Chooses Profits over Safety.'" *The New York Times*, October 3, 2021.

Sachdeva, Sam. . "Royal Society Investigation into Mātauranga Māori Letter Sparks Academic Debate." *Newsroom*. 17 November 2021.

Sanfilippo, Madelyn R. , Yan. Shvartzshnaider , Irwin. Reyes , Helen. Nissenbaum , and Serge. Egelman . "Disaster Privacy/Privacy Disaster." *Journal of the Association for Information Science and Technology* 71, no. 9

(2020): 1002–1014.

Sanfilippo, Madelyn. , Shengnan. Yang , and Pnina. Fichman . “Managing Online Trolling: From Deviant to Social and Political Trolls.” Proceedings of the 50th Hawaii International Conference on System Sciences (2017). <https://doi.org/10.24251/HICSS.2017.219>.

Sara, Malm. , “Belgian Woman Blindly Drove 900 Miles across Europe as She Followed Broken GPS instead of 38-Miles to the Station.” Daily Mail.com 14 January 2013. <https://www.dailymail.co.uk/news/article-2262149/Belgian-woman-67-picking-friend-railway-station-ends-Zagreb-900-miles-away-satnav-disaster.html>.

Sarah, Werner. . “Working towards a Feminist Printing History.” Printing History (2020).

Savolainen, Reijo. . “Everyday Life Information Seeking: Approaching Information Seeking in the Context of ‘Way of Life.’” Library & Information Science Research 17, no. 3 (1995): 259–294.

Schneider, Jodi. , Alexandre. Passant , and Stefan. Decker . “Deletion Discussions in Wikipedia: Decision Factors and Outcomes.” Proceedings of the Eighth Annual International Symposium on Wikis and Open Collaboration, edited by Cliff Lampe (Linz, Austria: Wikisym, 2012): 17.

Schwartz, Adam C. The Oracle Bone Inscriptions from Huayuanzhuang East. Berlin: De Gruyter, 2020.

“General Information.” Library of Congress.

<https://www.loc.gov/about/general-information/>. Retrieved 23 May 2022.

See Elle Hunt . “Tay, Microsoft’s AI Chatbot, Gets a Crash Course in Racism from Twitter.” The Guardian. 24 March 2016.

Selod, Saher. . “Gendered Racialization: Muslim American Men and Women’s Encounters with Racialized Surveillance.” Ethnic and Racial Studies 42, no. 4 (2019): 552–569.

Serres, Alexandre. . “Hypertexte: une histoire à revisiter.”

Documentaliste:Sciences de l’Information no. 32(2) (1995): 2.

Shackelford, Laura. , Wenhao David. Huang , Alan. Craig , Cameron. Merrill , and Danying. Chen . “Relationships between Motivational Support and Game Features in a Game-Based Virtual Reality Learning Environment for Teaching Introductory Archaeology.” Educational Media International 56, no. 3 (2019): 183–200.

Shane, Harris. and Dan. Lamothe . “Intelligence-Sharing with Ukraine Designed to Prevent Wider War.” Washington Post, 11 May 2022.

Šimić, Goran. . “To Believe or Not to Believe: Current History Textbooks in Bosnia and Herzegovina.” In Nationhood and Politicization of History in School Textbooks, edited by Gorana Ognjenović and Jasna Jozelić. Cham: Palgrave Macmillan, 2020.

Simone, Tom. and Gian M. Volpicelli . “Ukraine’s Digital Ministry is a Formidable War Machine.” Wired, 17 March 2022.

<https://www.wired.co.uk/article/ukraine-digital-ministry-war>.

Simpson, Rosemary. , Allen. Renear , Elli. Mylonas , and Andries. van Dam . “50 Years after ‘As We May Think’ the Brown/MIT Vannevar Bush Symposium.” Interactions 3, no. 2 (1996): 47–67.

Singer, Carol A. “Ready Reference Collections: A History.” Reference & User Services Quarterly, 49, no. 3 (2010): 253–264.

Singhal, Amit. . "Introducing the Knowledge Graph: Things, Not Strings. 16 May 2012. <https://blog.google/products/search/introducing-knowledge-graph-things-not/>.

Smale, Aaron. . "“Ko au te awa, ko te awa ko au.”" New Zealand Radio. 17 March 2017.

Small, Scott A. "Age-Related Memory Decline: Current Concepts and Future Directions." *Archives of neurology* 58, no. 3 (2001): 360–364.

Smith, Linda C. "Citation Analysis." *Library Trends* Summer (1981): 93.

Smith, Linda C. "“Memex as an Image of Potentiality in Information Retrieval Research and Development.”" In *Proceedings of the 3rd Annual ACM Conference on Research and Development in Information Retrieval*. Kent, UK: Butterworth (1980): 345–369.

Smith, Linda C. "Memex as an Image of Potentiality Revisited." In *From Memex to hypertext: Vannevar Bush and the Mind's Machine*, edited by James. Nyce and Paul. Kahn . London: Academic Press, 1991: 265.

Smith, Linda C. . *Procognitive Systems: J.C.R. Licklider's Vision for Library Systems of the Future*. University of Illinois, 2015.

Soranno, Patricia A. , Kendra S. Cheruvilil , Kevin C. Elliott , and Georgina M. Montgomery . "It's Good to Share: Why Environmental Scientists' Ethics are out of Date." *BioScience* 65, no. 1 (2015): 69–73.

Srivastava, Anugrah. , and Advait. Naik . "Big Data Analysis in Bioinformatics." In *Advances in Bioinformatics*, edited by Vijai. Singh and Ajay. Kumar . Singapore: Springer, (2021): 405–429.

Stafford, Pauline. , "Edith, Edward's Wife and Queen." In *Edward the Confessor: The Man and the Legend*, edited by Richard. Mortimer . Woodbridge: The Boydell Press, 2009, 126–128.

Star, Susan Leigh. and James. Griesemer . "Institutional Ecology, 'Translations' and Boundary Objects: Amateurs and Professionals in Berkeley's Museum of Vertebrate Zoology." *Social Studies of Science* 19 (3) (1989): 387–420.

Stenson, Molly Wright. . *Architectural Intelligence: How Designers and Architects Created the Digital Landscape*. Cambridge, MA: MIT Press, 2017.

Stephenson, Heather. . "Preserving Ukraine's Cultural Heritage Online." *Tufts Now*, 22 March 2022.

Stewart, Brenton. , and Boryung. Ju . "On Black Wikipedians: Motivations behind Content Contribution." *Information Processing & Management* 57, no. 3 (2020): 102134.

Stilgoe, Jack. . "How Can We Know a Self-Driving Car Is Safe?" *Ethics and Information Technology* 23, no. 4 (2021): 635–647.

Stone, Christopher D. "Should Trees Have Standing? Towards Legal Rights for Natural Objects." *Southern California Law Review* 45 (1972): 450–501.

Stromgren, Pip. . "Charles Ammi Cutter: Library Systematizer Extraordinaire." *Daily Hampshire Gazette*, 26 June 2004. <https://forbeslibrary.org/info/library-history/charles-ammi-cutter/>.

Suárez-Gonzalo, Sara. , Lluís Mas. Manchón , and Frederic Guerrero. Solé . "Tay Is You: The Attribution of Responsibility in the Algorithmic Culture." *Observatorio* 13, no. 2 (2019): 1–14.

Suárez-Gonzalo, Sara. . "Deadbots' Can Speak for You after Your Death. Is That Ethical?" The Conversation, 9 May 2022.

Sula, Chris Alen. , and Heather V. Hill . "The Early History of Digital Humanities: An Analysis of Computers and the Humanities (1966–2004) and Literary and Linguistic Computing (1986–2004)." Digital Scholarship in the Humanities 34, no. Supplement 1 (2019): i190–i206.

Tan, Shin Bin. , Colleen. Chiu-Shee , and Fábio. Duarte . "From SARS to COVID-19: Digital Infrastructures of Surveillance and Segregation in Exceptional Times." Cities 120 (2022).
<https://doi.org/10.1016/j.cities.2021.103486>.

"Text of the Convention for the Safeguarding of the Intangible Cultural Heritage." UNESCO Intangible Cultural Heritage.
<https://ich.unesco.org/en/convention>.

Thornton, Russell. . "Who Owns Our Past? The Repatriation of Native American Human Remains and Cultural Objects." In Native American Voices, edited by Susan. Lobo , Steve. Talbot and Traci L. Morris . New York: Routledge, 2016: 311.

Tiku, Nitasha. . "The Google Engineer Who Thinks The Company's AI Has Come to Life." Retrieved 12 June 2022.
<https://www.washingtonpost.com/technology/2022/06/11/google-ai-lambda-blake-lemoine/>.

"Timbuktu's 'Badass Librarians': Checking Out Books under Al-Qaida's Nose." In All Things Considered, Produced by National Public Radio, 23 April 2016. <https://www.npr.org/2016/04/23/475420855/timbuktus-badass-librarians-checking-out-books-under-al-qaidas-nose>.

Todorova-Ekmekci, Mirena. , Todor. Todorov , and Kalina. Sotirova-Valkova . "Usage of Innovative Technologies and Online Media Tools for Digital Presentation of Cultural Heritage in Bulgaria." Digital Presentation and Preservation of Cultural and Scientific Heritage 11 (2021): 303–308.

Toscano, Chloé Valentine. . "Go Ahead and Stare at My Prosthetic Arm. I Know It's Awesome." Washington Post, 19 September 2022.
<https://www.washingtonpost.com/lifestyle/2022/09/19/prosthetic-arm-design-personalize-disability/>. Retrieved 22 September 2022.

"Towards the Information Age: Paul Otlet (1868–1944) Founder of the Mundaneum." Google Arts & Culture. Retrieved 31 May 2022.
<https://artsandculture.google.com/story/awXRg4ha0wAA8A?hl=en>.

Trimmis, Konstantinos Prokopios. . "The Forgotten Pioneer: Valerios Stais and His Research in Kythera, Antikythera and Thessaly." Bulletin of the History of Archaeology 26, no. 1 (2016).

Tripodi, Francesca. . "Ms. Categorized: Gender, Notability, and Inequality on Wikipedia." New Media & Society (June 2021): 1–21.
<http://doi.org/10.1177/14614448211023772>.

Trompf, Gary W. "The Classification of the Sciences and the Quest for Interdisciplinarity: A Brief History of Ideas from Ancient Philosophy to Contemporary Environmental Science." Environmental Conservation 38, no. 2 (2011): 113–126.

Turk, Matthew J. "Scaling a Code in the Human Dimension." In Proceedings of the Conference on Extreme Science and Engineering Discovery Environment: Gateway to Discovery, 2013: 1–7.

<https://doi.org/10.1145/2484762.2484782>

Tynan, Lauren. . "What Is Relationality? Indigenous Knowledges, Practices and Responsibilities with Kin." *Cultural Geographies* 28, no. 4 (2021): 597–610.

Underwood, Ted. . "A Genealogy of Distant Reading." *DHQ: Digital Humanities Quarterly* 11, no. 2 (2017).

<http://www.digitalhumanities.org/dhq/vol/11/2/000317/000317.html>.

Retrieved 22 September, 2022.

Universal Declaration of Human Rights, United Nations (1948).

<https://www.un.org/en/about-us/universal-declaration-of-human-rights>.

Van den Heuvel, Charles. . "Web 2.0 and the Semantic Web in Research from a Historical Perspective: The Designs of Paul Otlet (1868-1944) for Telecommunication and Machine Readable Documentation to Organize Research and Society." *Knowledge Organization* 36, no. 4 (2009): 214–226.

Vick, Karl. . "Bellingcat's Eliot Higgins Explains Why Ukraine Is Winning the Information War." *Time*. 9 March 2022. <https://time.com/6155869/bellingcat-eliot-higgins-ukraine-open-source-intelligence/>.

Virkus, Sirje. , and Emmanouel. Garoufallou . "Data Science and Its Relationship to Library and Information Science: A Content Analysis." *Data Technologies and Applications* 54, no. 5 (2020): 643–663.

"Vivian G. Harsh Research Collection." Chicago Public Library. Retrieved 11 June 2022. <https://www.chipublib.org/vivian-g-harsh-research-collection/>.

Vogele, Jessica. . "Where's the Fair Use: The Takedown of Let's Play and Reaction Videos on YouTube and the Need for Comprehensive DMCA Reform." *Touro Law Review* 33 (2017): 589.

von Oswald, Margareta. . "The 'Restitution Report': First Reactions in Academia, Museums, and Politics." Centre for Anthropological Research on Museums and Heritage (2019). Retrieved 30 October 2022. <https://blog.uni-koeln.de/gssc-humboldt/the-restitution-report/>.

Wachs, Johannes. , and Balázs. Vedres . "Does Crowdfunding Really Foster Innovation? Evidence from the Board Game Industry." *Technological Forecasting and Social Change* 168 (2021): 120747.

Wagner, Travis L. , and Ashley. Blewer . "The Word Real Is No Longer Real: Deepfakes, Gender, and the Challenges of AI-Altered Video." *Open Information Science* 3, no. 1 (2019): 32–46.

Walker, Christopher. , and Ann. Copeland . "The Eye Prophetic: Julia Pettee." *Libraries & the Cultural Record* 44, no. 2 (2009): 166.

Wamsley, Laurel. . "Library of Congress Will No Longer Archive Every Tweet." *NPR*, 26 December 2017.

Wang, Tian. , Lin. Guo , and Masooda. Bashir . "COVID-19 Apps and Privacy Protections from Users' Perspective." *Proceedings of the Association for Information Science and Technology* 58, no. 1 (2021): 357–365.

Wang, Yang. . "First-Time Internet Users in Nigeria Use the Internet in a Unique Way." *KaiOS Industry Insights* (2020).

<https://www.kaiostech.com/first-time-internet-users-in-nigeria-use-the-internet-in-a-unique-way-heres-why-that-matters/>.

Wang, Yang. . "Inclusive Security and Privacy." *Sociotechnical Security and Privacy. IEEE Computer and Reliability Societies*, July/August 2018: 82–87.

Watts, Duncan. and Steven H. Strogatz . "Collective Dynamics of 'Small-World' Networks." *Nature* 393, no. 6684 (1998): 440–442.

Weinberger, David. . "The Problem with the Data-Information-Knowledge-Wisdom Hierarchy." *Harvard Business Review* 2 (2010).
<https://hbr.org/2010/02/data-is-to-info-as-info-is-not>.

Weinstein, Emily. . "The Social Media See-Saw: Positive and Negative Influences on Adolescents' Affective Well-Being." *New Media & Society* 20, no. 10 (2018): 3597–3623.

Wells, Herbert George. . *World Brain*. Cambridge, MA: MIT Press, 2021.

Westengen, Ola T. , Charlotte. Lusty , Mariana. Yazbek , Ahmed. Amri , and Åsmund. Asdal . "Safeguarding a Global Seed Heritage from Syria to Svalbard." *Nature Plants* 6, no. 11 (2020): 1311–1317.

"What Is GDPR, the EU's New Data Protection Law?" GDPR.EU. Retrieved 6 June 2022. <https://gdpr.eu/what-is-gdpr/>.

Whelan, Allison M. "Unequal Representation: Women in Clinical Research." *Cornell Law Review Online* 106 (2020): 103–104.

Wiegand, Wayne A. "Sanitizing American Library History: Reflections of a Library Historian." *The Library Quarterly* 90, no. 2 (2020): 108–120.

"Wikipedia." Wikipedia. Retrieved 29 May 2022.
<https://en.wikipedia.org/wiki/Wikipedia>.

"Wikipedia: Wiki Game." Retrieved 29 May 2022.
https://en.wikipedia.org/wiki/Wikipedia:Wiki_Game.

Wilson, Thomas D. "Human Information Behavior." *Informing Science* 3, no. 2 (2000): 50.

Witzel, Michael. . "Vedas and Upaniṣads." In *The Blackwell Companion to Hinduism*, edited by Gavin. Flood . Oxford and Malden, MA: Blackwell, 2003: 68–101.

Wolf, Gary. . "The Curse of Xanadu." *Wired*, 1 June 1995.

Woods, Christopher. . "The Earliest Mesopotamian Writing." In *Visible Language: Inventions of Writing in the Ancient Middle East and Beyond*. Chicago, IL: Oriental Institute Museum Publications, 2010: 32.

"World IA Day." Retrieved 12 June 2022. <https://www.worldiaday.org/>.

Wright, Alex. , "The Web Time Forgot." *The New York Times*. 17 June 2008.

Wright, Alex. . "The Future of the Web Is 100 Years Old." *Nautilus* (29 January 2015). Retrieved 2 June 2022. <https://nautil.us/the-future-of-the-web-is-100-years-old-2894/>.

Wright, Alex. . "The Secret History of Hypertext." *The Atlantic*, 22 May 2014.

Wright, Alex. . *Cataloging the World: Paul Otlet and the Birth of the Information Age*. Oxford: Oxford University Press, 2014.

Wright, Michael T. "Archimedes, Astronomy, and the Planetarium." In *Archimedes in the 21st Century*, edited by Chris. Rorres (Cham: Birkhäuser, 2017): 125–141.

Wright, Sarah. , Kate. Lloyd , Sandie. Suchet-Pearson , Laklak. Burarrwanga , Matalena. Tofa , and Bawaka. Country . "Telling Stories in, through and with Country: Engaging with Indigenous and More-Than-Human Methodologies at Bawaka, NE Australia." *Journal of Cultural Geography* 29, no. 1 (2012): 39–60.

Wu, Tim. . "Whatever Happened to Google Books." The New Yorker, 11 September 2015.

Yang, Dong. , Ziyue. Xu , Wenqi. Li , Andriy. Myronenko , Holger R. Roth , Stephanie. Harmon , Sheng. Xu et al. "Federated Semi-Supervised Learning for COVID Region Segmentation in chest CT Using Multi-National Data from China, Italy, Japan." Medical Image Analysis 70 (2021): 101992.

Yavuz, N. Kivılcım. . "Hernando Colón's Book of Books: AM 377 fol."

Retrieved 17 May 2022. <https://nkyavuz.com/blog/ams-am-377-fol/>.

"Yewno." Retrieved 1 June 2022. <https://www.yewno.com/about>.

Yong, Ed. , "This Speck of DNA Contains a Movie, a Computer Virus, and an Amazon Gift Card." The Atlantic. 2 March 2017.

Zargar, Safwat. . "Mistaken Identity: A Kashmiri Farmer on Pilgrimage in Iraq Has Been Detained on an Interpol Notice." Scroll.In, 12 January 2020.

<https://scroll.in/article/949613/mistaken-identity-a-kashmiri-farmer-on-pilgrimage-in-iraq-has-been-detained-on-an-interpol-notice>.

Zins, Chaim. . "Conceptions of Information Science." Journal of the American Society for Information Science and Technology 58, no. 3 (2007): 335–350.

Zipf, George Kingsley. . Human Behavior and the Principle of Least Effort: An Introduction to Human Ecology. Boston, MA: Addison Wesley Press, 1949.

"Zooniverse.org." Retrieved 3 June 2022 <https://www.zooniverse.org/about>.